

Gilian®



GilAir® PLUS

Operation Manual

SENSIDYNE®
Industrial Health & Safety Instrumentation

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REF 360-0132-01 (Rev D)

GilAir Plus Identifiers:



A LCD Display

B Status and Notification LED's

C Inlet Filter

D Hi/Lo Control Valve

E Charging Contacts

F Belt Clip

G Battery Access Screws

H Power Port

I USB Port

J Reference Device Port (On communication-enabled docks)

Quality Policy Statement

At Sensidyne, we are committed to providing products and services that consistently meet customer needs and comply with all applicable statutory and regulatory requirements.

Our products are designed and manufactured in accordance with ISO 9001:2008, EN 13980:2002, ATEX Directive 94/9/EEC, and IECEx. Through ongoing review of our designs, supplier performance, and customer feedback we strive to ensure continuous improvement.

All employees at Sensidyne share the responsibility to provide products that are produced efficiently and economically representing the best value to our customers. We are committed to meeting or exceeding customer expectations in everything we do.

Sensidyne, LP

Warranty

Sensidyne warrants that, at the time of delivery, the GilAir Plus shall be free of all defects in workmanship and material. Sensidyne will repair or replace, at its sole option, any GilAir Plus found to be defective by Sensidyne, if notified by Purchaser within the Warranty time period.

The warranty time period shall be for two (2) years from the date of original shipment by Sensidyne, except as noted below.

- A. Exceptions to the above two year warranty time period:
 - 1. The keypad of the GilAir Plus has a five (5) year warranty
 - 2. The rechargeable NiMH battery pack has a one (1) year warranty.
 - 3. Consumables have a ninety (90) day warranty.

- B. This warranty shall be null and void on any product which:
 - 1. Is operated or used in excess of the product's operating specifications; or
 - 2. is not properly maintained in accordance with its maintenance manual or specifications; or
 - 3. has been repaired or modified by persons other than authorized Sensidyne personnel or Factory Trained Service Centers, unless such work is authorized in advance in writing by Sensidyne; or
 - 4. has been damaged, abused, or misused.

- C. Warranty on Service and Repairs:
 - 1. Goods, which have been repaired or replaced during the warranty period, are warranted only for the remainder of the unexpired portion of the original warranty period.
 - 2. Repairs or service provided not pursuant to warranty: 180 days from date of shipment by Sensidyne.

THIS WARRANTY IS IN LIEU OF ALL OTHER WARRANTIES, EXPRESSED OR IMPLIED, INCLUDING BUT NOT BEING LIMITED TO THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR USE FOR A PARTICULAR PURPOSE, WHICH ARE EXPRESSLY DISCLAIMED, AND CONSTITUTES THE ONLY WARRANTY OF SENSIDYNE WITH RESPECT TO GOODS SOLD OR DELIVERED.

Table of Contents

GilAir Plus Identifiers.....	II
Quality Policy Statement	III
Warranty.....	IV
SECTION ONE: Preface	1
Warnings	3
Certifications, Approvals and Compliances	5
SECTION TWO: Introduction.....	7
2.1. Product Description	7
2.2. Pump Kit Descriptions	8
SECTION THREE: Set-Up	9
3.1. Preparation	9
3.2. Pump Start Up.....	9
3.2.1. Power Up	9
3.2.2. Idle Mode	9
3.2.3. Power Down.....	9
3.3. Setting the Flow Rate	10
3.4. Power Options.....	10
SECTION FOUR: General Operation.....	11
4.1. Overview	11
4.2. Connections	11
4.3. Navigation	12
4.4. Menus.....	12
4.5. Displays.....	14
4.5.1. Display Details	16
4.5.2. Menu Details	16
4.5.3. Idle Display Details.....	16
4.5.4. Constant Flow Run Display Details	17
4.5.5. Constant Pressure Display Details.....	17
4.5.6. Fault Display Details	17
4.5.7. Program Display.....	18

4.5.8. STP Display	18
4.6. Sensor Calibration	19
4.7. Run Mode	19
4.7.1. Run Mode Descriptions	19
4.7.2. Locking the Keypad	19
4.7.3. Unlocking the Keypad	20
4.8. Flow Set (cc/min)	20
4.8.1. Setting the Flow Rate Range	20
4.8.2. Setting the Flow Rate	22
4.9. Field Calibrate	22
4.9.1. Field Calibration	22
4.9.2. Calibration Option (SmartCal SM)	23
4.9.3. Displayed Flow Calibration	23
4.10. Causes of Fault Condition	25
SECTION FIVE: Options.....	27
5.1. Setup ►	27
5.2. Event ID Enable.....	27
5.3. Pre/Post-Calibration	28
5.4. Fault Retry	29
5.5. Valve Mode	30
5.6. SmartCal SM Automatic Calibration	30
5.6.1. Gilibrator-2	31
5.6.2. Challenger [®]	32
5.6.3. TSI [™] Model 4140	33
5.6.4. Bios Defender [™] 510	34
5.7. Clear Datalog.....	35
5.8. Run Options ►	36
5.8.1. Standard Temperature (STP Models only).....	36
5.8.2. Standard Pressure (mmHg) (STP models only)	36
5.9. Display Options ►	37
5.9.1. Language	37
5.9.2. Temperature Units	37
5.9.3. Pressure Units.....	38
5.10. Clock Set ►	38

5.10.1. Clock	38
5.10.2. Date	39
5.10.3. Time Format.....	39
5.10.4. Date Format	40
5.11. Password.....	40
5.12. Control Mode	41
5.13. Run Mode Manual, Timed, Program Name	42
5.14. Run Setup ►	43
5.14.1. Timed Start.....	43
5.15. Timed Duration.....	43
5.16. Pressure Set ("H ₂ O or mmHg or KPa).....	44
SECTION SIX: Programming	45
6.1. Program ►	45
6.2. Enabling the Program Mode	45
6.3. Program Edit ►	46
6.4. Program Name	46
6.5. Control Mode	47
6.6. Set the Flow Rate	47
6.7. Program Steps.....	48
6.8. Function.....	48
6.9. Function Value.....	49
6.10. Save a Program.....	50
6.11. Review Events.....	50
SECTION SEVEN: PC Interface	51
7.1. PC Interface	51
SECTION EIGHT: Maintenance Menu	52
8.1. Maintenance ►	52
8.2. Global Reset.....	52
8.2.1. Reset (save programs).....	54
8.3. T ambient Cal ►	54
8.4. Barometric P Cal ►	55
8.5. Pressure ►	56

SECTION NINE: User Maintenance	58
9.1. Battery Maintenance NiMH.....	58
9.2. Battery Replacement	58
9.3. Pump Filter Maintenance.....	59
SECTION TEN: Appendices.....	60
Appendix A: Menu Outline.....	60
Appendix B: Example Program Setup & Edit	62
Appendix C: Troubleshooting Guide	66
Appendix D: Parts List.....	69
Appendix E: Specifications.....	73
Appendix F: Charging/Communications Dock.....	75
Appendix G: Factory Calibration and Service.....	77

SECTION ONE: Preface

Proprietary Notice

The intended use of this manual is exclusive to owners of Gilian GilAir Plus air sampling pumps. The material within this manual is proprietary information and is to be used only to understand, operate, and service the instrument. By receiving this document, the recipient agrees that neither this document, the information disclosed within, nor any part thereof shall be reproduced or transferred, physically, electronically or in any other form or used or disclosed to others for manufacturing or for any other purpose except as specifically authorized in writing by Sensidyne, LP.

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WARNINGS

READ AND UNDERSTAND ALL WARNINGS AND INSTRUCTIONS BEFORE USE.

Failure to read, understand, and comply with ALL accompanying literature, instructions, product labels, and warnings could result in property damage, severe personal injury, or death.

Read and understand ALL applicable environmental health and safety laws and regulations before operating this product. Ensure complete compliance with ALL applicable laws and regulations before and during the use of this product.

DO NOT remove, cover, or alter any label or tag on this product, its accessories, or related products.

UNDER NO CIRCUMSTANCES should this product be used except by qualified, trained and technically competent personnel.

The GilAir Plus portable air sampling pump is intended for both indoor and outdoor use. The unit is not waterproof. NEVER submerge the unit in water. Pump failure, faulting or user injury may result.

The GilAir Plus Pump is Intrinsically Safe when used with specified battery pack part number 783-0012-01-R. Refer to the Certifications and Approvals section for approval ratings. **Due to risk of static charge, do not clean the pump labels or keypad with a dry cloth in areas where acetylene may be present.**

DO NOT operate this product should it malfunction, require repair, or have a cracked or broken case or other visible or known damage.

DO NOT repair or modify this product, except as specified in this Operation Manual. All user controls and adjustments are made via the sealed keypad on the front of the pump and the Hi/Lo control valve. The only user-replaceable parts are the Battery Pack and Pump Filter. (See Sections 9.2. and 9.3.)

Use ONLY specified Sensidyne parts when performing maintenance procedures described in this manual. Intrinsic safety certifications become void by substitution of unauthorized components, unauthorized repair or alteration. All other Service should be performed by Sensidyne Authorized Service Departments only. (See Appendix D for Parts List; see Appendix G for Service Contact Information).

This product uses rechargeable Nickel Metal Hydride (NiMH) batteries. Always fully charge before use. **DO NOT open the pump case, charge or replace batteries in an explosive atmosphere.** Use only battery pack and chargers specified in the Parts List. DO NOT operate pump while charging. **Caution:** Both charger and battery may become warm during charging.

This product offers an optional battery configuration that will accept over-the-counter alkaline, lithium, or rechargeable NiMH batteries. **The GilAir Plus is not intrinsically safe when used in this configuration and should not be used in explosive atmospheres when using this optional battery configuration.**

If the GilAir Plus pump comes into contact with a destructive substance(s) it is the responsibility of the user to take suitable precautions that prevent the pump from being adversely affected, thus ensuring that the Intrinsic Safety protection is not compromised. Destructive substances include acidic liquids or gases that may attack metals, solvents that may affect polymeric materials, other solvents, or corrosives. Suitable precautions are regular checks as part of routine inspections and establishing from material data sheets that chemicals known to be present do not have an adverse effect on the material of the pump (polycarbonate, polyester, silicone, Buna-N, Neoprene, Stainless steel, brass and epoxy).

DO NOT operate with a dirty or blocked inlet filter or kinked tubing. Pump failure or faulting may result.

Certifications, Approvals and Compliances

Declaration of Conformity

DECLARATION OF CONFORMITY

Sensidyne, LP

16333 Bay Vista Drive Clearwater, Florida 33760 U.S.A.

Certificate No: 001

Issue 2 June 1, 2011

The undersigned declares that the products named in this certificate meet the provisions of the European Communities Council Directive 94/9/EC (Atex) concerning equipment and protective systems intended for use in potentially explosive atmospheres and US and Canadian Hazardous Location and Electrical Equipment Requirements.

Product Type: Portable Air Sampling Pump

Product designation: Gilair Plus with 7.2 V DC NiMH Battery Pack

Manufacturer : Sensidyne, LP
16333 Bay Vista Drive
Clearwater Florida 33760, USA

Intended Use: Air Sampling

Notified Body: FM Approvals Ltd.
1 Windsor Dials
Windsor
Berkshire
UK SL4 1RS

Notified body Number; CE 1725



Intrinsically Safe:

Class I, Division 1, Groups: A, B, C, D, E, F, and G, hazardous (classified) locations.

IS/I,II,III /1/ ABCDEFG / T4 ta - 0°C to 45°C

Class 1, Zone 0, Group: IIC hazardous (classified) locations.

1 / 0 AEx ia IIC /T4 Ta - 0°C to 45°C

IECEX FMG 10.0019X

Ex ia IIC T4 Ga Ta - 0°C to 45°C

Conforming to the following standards:

IEC 60079-0: 2007-10 Edition 5

IEC 60079-11:2006 Edition 5

IEC 60079-26:2006 Edition 2

Test Report: US/FMG/ExTR10.0023/00

QAR: GB/SIR/QAR08:0026/01

EC FM10ATEX0044X

II 1 G Ex ia IIC T4 Ta - 0°C to 45°C

Conforming to the following standards:

EN60079-0:2006

EN60079-11:2007

EN60079-26:2007

Report: 3039791EC

ATEX Quality Assurance Notification: SIRA Certification Notified body Number: 0518

GilAir Plus 001

EMC: Emissions and Immunity Standards
EN 61326:2006 EN 55011: Class A
EN 61000-4-2
EN 61000-4-3
EN 61000-4-4
EN 61000-4-5
EN 61000-4-6
EN 61000-4-8
EN 61000-4-11
Reference Product Safety Engineering Report 10162

Safety Compliance:
TUV NRTL: U8 11 03 71335-002
CSA/CAN C22.2 No. 61010-1-04
UL 61010-1:2001:2005
EN 61010-1
Report: 090-1002842-000

Additional Standards:
EN1232

Signed:



Date: 6-1-11

Title: Quality Assurance and Regulatory Affairs Manager Sensidyne, LP

Who is the natural and legal person with responsibility for the design, manufacture, packaging and labeling before the device is placed on the market under his own name, regardless of whether these operations are carried out by the Manufacturer or on his behalf by a third party.

GilAir Plus 001

SECTION TWO: Introduction

2.1. Product Description

The GilAir Plus is an advanced personal air sampling pump system available in three models: a basic model, a datalogging pump (DL) model, and standard temperature and pressure (STP) model. The STP model corrects the display flow rate and air volume to standard conditions of temperature and pressure.

Pump Model	Flow Rate 1 to 5000 cc/min	Constant Flow & Constant Pressure	Battery Options NiMH, Alkaline & DC	Simple Program Functions (Timer Only)	Advanced Program Functions	Datalogging & Transfer to PC	Automatic Calibration Option (SmartCal)	Standard Temp & Pressure Data
Basic	✓	✓	✓	✓			✓*	
Datalog	✓	✓	✓	✓	✓	✓	✓	
STP	✓	✓	✓	✓	✓	✓	✓	✓

All models offer constant flow modes and constant pressure control modes. The Constant Flow Mode holds the set flow constant against changing back pressures within 5% or 3 cc/min whichever is larger. The Constant Pressure Mode holds the pressure constant when taking samples using constant resistance sampling trains. The constant pressure (multi-flow) mode allows the air stream to be split into two or more separate samples so that multiple samples can run at the same time. Furthermore, all models offer a built-in low flow mode, such that flow rates are attainable from 1 cc/min to 5000 cc/min without the addition of an external low flow adapter. Additionally, all models offer start-stop programmability. A matrix chart for the available pump models is shown above.

***Note:** SmartCalSM Automatic calibration is only available on all pump models when used with a communications-enabled Dock. The **standard** dock/charger that comes with the basic pump does not have communication capability and will not support the SmartCal feature.

2.2. Pump Kit Descriptions

Kits are available in one, three and five pump configurations, with power cords in US, Euro and UK versions. A matrix chart for the available kits is provided below.

Kit Type	GilAir Plus Pump	Dock	Carrying Case	Filter Cassette Holder	Charcoal Tube Holder	CD with Software and Manual / Printed Manual
Single Starter Kit						
Basic	1	Single-Station Standard	No	1	1	1/1
Datalog (DL)	1	Single-Station with Communication	No	1	1	1/1
STP	1	Single-Station with Communication	No	1	1	1/1
Three-Pack Starter Kit						
Basic	3	Three-Station Standard	Yes	3	3	1/1
Datalog (DL)	3	Three- Station with Communication	Yes	3	3	1/1
STP	3	Three- Station with Communication	Yes	3	3	1/1
Five-Pack Starter Kit						
Basic	5	Five- Station Standard	Yes	5	5	1/1
Datalog (DL)	5	Five- Station with Communication	Yes	5	5	1/1
STP	5	Five- Station with Communication	Yes	5	5	1/1

See Appendix D for part numbers for pumps, kits and accessories.

SECTION THREE: Set-Up

3.1. Preparation

The package includes the pump, dock, dock power supply, and a line cord. The dock serves as the charging base for all models and the communications dock for the datalogging and STP models.

Plug the power supply into the dock and the AC power cord into the power supply. Connect the AC power cord to mains supply. The supply can accept 100-240vAC, at 50 or 60 Hz.

The pump arrives fully assembled.

Charge battery to full capacity before using the pump. To charge the pump, place it onto the charging base. The pump's belt clip secures the pump in place. Connection is made via contact points on both sides of the belt clip.

Allow up to three and one-half hours for a complete charge. A red LED on the pump indicates charging in progress; a green LED indicates charged and ready for use. The green LED will flash during top off charge and is on constantly while on trickle charge.

3.2. Pump Start Up

3.2.1. Power Up

Power pump on by pressing and holding down (about 2 seconds) the  button until the pump turns on.

3.2.2. Idle Mode

Pump will enter idle mode for five seconds then display the main menu.

3.2.3. Power Down

Turn power off from any display when the pump is not running by pressing and holding down the  button. After approximately two seconds a power down window will appear and a five second shutdown sequence will start. If the button is released before the shutdown sequence is complete the power will not switch off. At the termination of the shutdown sequence, power is off.

Note: Power cannot be shut down if a sample or program is running.

3.3. Setting the Flow Rate

On the main menu, use the  button to move the cursor to **Flow Set**. Adjust the flow rate to the desired value using the  and  buttons. Press and release the  button to confirm the change.

3.4. Power Options

The GilAir Plus comes standard with a rechargeable nickel metal hydride (NiMH) battery. The unit is charged through the dock that is included with all starter kits. An optional alkaline battery pack (P/N 783-0013-01-R) is available that allows the use of over-the-counter alkaline or lithium batteries. A third option, the DC power adapter (783-0014-01-R) allows extended run times with the pump on the dock.

Note: Do not attempt PC communications with the Dock, when running a sample using the DC power adapter or alkaline cell battery pack.

Warning: *Alkaline batteries, lithium batteries and the DC power adapter should only be used in non-hazardous areas. Intrinsic safety certifications are only valid when using the rechargeable NiMH battery pack.*

SECTION FOUR: General Operation

4.1. Overview

The GilAir Plus has the capability of generating and controlling flow over the range of 1 cc/min to 5000 cc/min in two flow ranges, 1-449 cc/min, and 450-5000 cc/min, that are selectable using a 2 mm or 5/64 inch hex key (provided with the pump). The actual flow is measured and controlled by the pump's internal processor. Flow control is provided directly in the constant flow mode. Pressure control is provided in the constant pressure control mode, which controls flow indirectly. During a sampling event the flow rate is displayed in the constant flow mode and back pressure is displayed in the constant pressure mode. Pump flow is not displayed in the constant pressure mode.

4.2. Connections

Sample media is connected to the inlet port using ¼ inch ID tubing. Adapters that cause high pressure drop or using smaller diameter tubing may affect the sample flow. Minimize pressure drop in tubing and fittings and avoid any condition that will exceed the pump back pressure specifications (see Appendix E). The input port is part of the input manifold that provides input connection, output connection and contains a filter that protects the pump from contamination by particulates if operated without an effective sampling filter. This filter is user replaceable and should be replaced if discolored, clogged or obstructed in any way.

An output adapter (included with pump kits) accommodates filling sample containers such as Tedlar or Kynar sampling bags. Connect the bag fill adapter and the sample bag using low-resistance tubing. If the pressure in the bag increases, such as when it is completely filled, the pressure increase will be shown as increased pump back pressure and will terminate the event if the back pressure specification is exceeded.

4.3. Navigation

The GilAir Plus pump uses an intuitive six-button keypad for menu navigation and pump operation. The buttons and their functions are summarized in the table below.

Symbol	Name	Function
	Power/Enter	Powers the unit up or down, and enters a menu or confirms a parameter change
	Escape	Exits a menu
	Up Arrow	Select display screen options or move cursor up menu
	Down Arrow	Select display screen options or move cursor down menu
	Increase	Adjusts a parameter to higher values
	Decrease	Adjusts a parameter to lower values

4.4. Menus

The operation of the pump is controlled by entering the menu system and selecting the menu item that performs the desired function. The menu has submenus that allow control of related functions. A quick reference menu outline appears in Appendix A.

When the menu is not displayed, the  and  buttons select among the display screen options. The display screens are automatically selected by the pump whenever the mode of operation is changed (example: when an event is started or a fault occurs). The display screens include the **Idle** display, the **Constant Flow** control operating display, the **Constant Pressure** control operating display, the **Fault** display, **Program Status** display and **STP Display** (STP Models only). A description of the data displayed on each menu screen is contained in the Displays section (Section 4.5.) of this manual.

Sampling events: The GilAir Plus accumulates sample data on events. An event is a sample run. The data for each sample event is stored as the event progresses so that no data loss is possible. When a sample is started, the event data is cleared to begin the new event. As the pump samples, the flow is controlled at the event flow rate or pressure depending on the selected control mode. The event data (time, flow, volume, backpressure and other information) is stored into nonvolatile review memory.

If the pump is paused the event is not terminated. When the event continues, volume and time accumulation continue in the same event. If the event is stopped, the run data is available for review (Section 6.11.) and in the idle screen until another event is started. The data from the last 16 events is available in the data review menu.

General operation of the menu system: Access the menu by pressing the  button. The menu is displayed in a vertical column. Some menu items have a ► sign after the menu item and lead to submenus. When the selected menu has the ►, pressing the  button enters the submenu. Pressing the  button exits the submenu and returns to the higher level menu. If a parameter is displayed at the right end of the line, the  and  buttons may be used to adjust the parameter up and down. If a parameter is changed it will be retained only if the  button is pressed to confirm the change. If a parameter has been changed, an ▲ symbol (for an increase in the parameter), or a ▼ symbol (signifying a decrease) is displayed to alert that the  button is required to confirm the change.

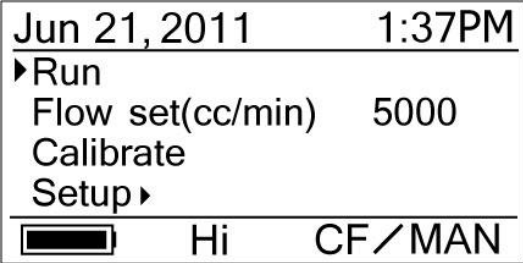
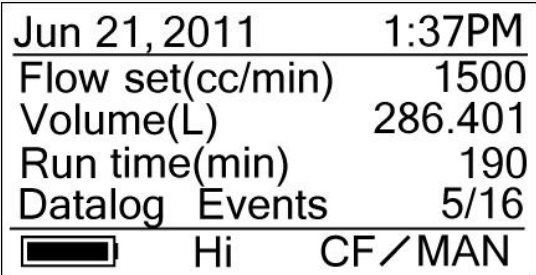
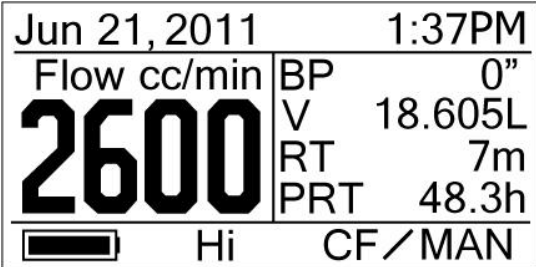
If the menu selection is altered with the  or  buttons without confirmation with the  button, the parameter will revert to its original value.

If the value is numeric, the  and  buttons alter the numeric value. If the parameter is a selection, the buttons will select each option in sequence.

NOTE: Many parameters have numeric values, you can scroll through the entire range of numeric values by simply pressing and holding down either the  and  buttons. Additionally, pressing and holding *first* the  button while then pressing and releasing the up arrow  button will automatically advance the parameter to the highest numeric value. Likewise, pressing and holding *first* the  button, and then pressing and releasing the  button will automatically advance the parameter to the lowest numeric value.

4.5. Displays

The display screens include the **Menu** display, **Idle** display, **Constant Flow** control operating display, **Constant Pressure** control operating display, **Fault** display, and the **Program Status** display. All displays include; the top status line displaying the Date, Time, and lock status; bottom status line displays the Battery charge status, Operating range (Hi or Lo), Control Mode (CF, CPL or CPH), and Run Mode (Manual, Timed, or Selected Program).

Displays	Description
Menu Display 	The main menu allows all of the parameters of the operation of the GilAir Plus to be set, by scrolling to the appropriate menu item (Section 4.3.)
Idle 	Shown when the pump is not conducting an event, Idle displays Flow Set Point, Total Sample Volume, Run Time and the number of used event data slots and the total number of slots.
Constant Flow 	Displays Actual Flow rate (cc/min), Back pressure BP (inches water, mmHg or KPa), Total Sample Volume V (Liters), Run Time RT (minutes) and Projected Run Time PRT (hours), based on battery charge status and back pressure.

Constant Pressure Jun 21, 2011 1:37PM BP "H2O 18.0 RT 8m PRT 48.3h Hi CPH/MAN	Displays Back pressure BP (inches water, mmHg, KPa), Run Time RT (minutes) and Projected Run Time PRT (hours) based on battery charge status and back pressure.
Fault Jun 21, 2011 12:37PM Fault RT: 1m FC: 1 FT: 0s Current: Flow > 5% Last: Pressure > max Lo CF/MAN	Displays Run Time RT (min), Fault Count FC, Fault Run Time FT (sec), Current Fault and Last Fault when a fault condition is detected or when the pump has stopped due to a fault condition.
Program Status Jun 21, 2011 12:31PM Program: PROG 1 Steps: 1 1: On 1m/5m Next: End Hi CF/PROG	Displays the status of a programmed run. Number of steps in the total program and the current and next step are shown. In the current step the progress of the step is shown.
STP Jun 21, 2011 1:47PM Ta 25(°C) Fa 4000cc Ts 25(°C) Fs 4038cc Pa 776mmHg Va 23.264L Ps 740mmHg Vs 24.384L Hi CF/MAN	In the STP model pump, this screen shows ambient temperature and pressure and the standard values. Flow and Volume are shown at ambient and standard conditions.

The LCD display can be switched between several information screens through the keypad. The pump will automatically switch to the appropriate display when certain menu items are selected or the status of the pump changes.

4.5.1. Display Details

1. Date and time: Displayed in selectable format mm/dd/yy or dd/mm/yy
2. Battery icon: When using a NiMH battery pack the charge state of the battery is shown. Charge status is approximate and depends on many factors including age, number of charge cycles, temperature and recent charge or discharge history.
3. Flow range indicator: Displays "Hi", "Lo" or "Err" at all times. This indicator shows the setting of the flow control valve, which determines high range or low range operation of the pump. When the indicator shows "Err", the valve is between positions and must be moved to the correct detent position to operate the pump.
4. Control mode: Displays the event control mode of the pump, either constant flow (CF) or constant pressure (CPH or CPL). In constant flow mode, the pump controls the flow rate regardless of changes in the filter load (back pressure). In constant pressure, the pump controls the inlet pressure, regardless of the flow. The constant pressure modes, CPH and CPL, allow the user to select the range of expected flow for optimum control.
5. Run mode: Displays the status of the Run mode, which indicates how the pump's sampling schedule is set. There are three modes; manual, the operator turns the pump on and off; timed, the pump turns on at a preset time for a preset duration; and program, indicated by the name of the program, operating under the timing control of a user defined program which specifies on and off times and durations.

4.5.2. Menu Details

1. Menu items: See the Menu Outline in Appendix A for an outline of the pump menu structure.

4.5.3. Idle Display Details

Idle Display: The idle display is visible when the pump is turned on and whenever the pump is not in the menu or a sample event.

1. Flow set (cc/min): The flow rate, always shown in cc/min at ambient conditions.
2. Volume (L) - Sample volume: The total volume of the sample, always shown in liters, at ambient conditions.
3. Run time (min) – Sample time in minutes.

4. Datalog Events: Displays the number of events recorded, and the total number of storage positions.

4.5.4. Constant Flow Run Display Details

1. Flow cc/min: The flow rate, always shown in cc/min at ambient conditions.
2. BP - Back pressure: The back pressure measured at the input to the pump after the manifold and protective filter. The units of the back pressure can be selected by the operator as inches of water, millimeters of mercury, or kilopascals.
3. V - Sample volume: The total volume of the sample, always shown in liters, at ambient conditions.
4. RT - Run time: Total pump **ON** time. Does not include Sensor calibrations, pauses or scheduled off time in programs.
5. PRT - Projected run time: The projected run time is an estimate of the remaining run time in hours, based on the current estimated battery capacity and the current drain on the pump, which is dependent on flow rate, backpressure and temperature.

4.5.5. Constant Pressure Display Details

1. BP - Back pressure: The back pressure measured at the input to the pump after the manifold and protective filter. The units of the back pressure can be selected by the operator as inches of water, millimeters of mercury, or kilopascals.
2. RT - Run Time: Total pump **ON** time. Does not include Sensor calibrations, pauses or scheduled off time in programs.
3. PRT - Projected Run Time: The projected run time is an estimate of the remaining run time in hours, based on the current estimated battery capacity and the current drain on the pump, which is dependent on flow rate, backpressure and temperature.

4.5.6. Fault Display Details

1. RT - Run time: Total pump **ON** time. Does not include Sensor calibrations, pauses or scheduled off time in programs.
2. FC - Fault count: Total number of faults that have resulted in suspension of sampling. When the count reaches 10, the pump will cease retrying and terminate the event.

3. FT - Fault time: Total time, in seconds, the pump has run while being in a fault status.
4. Current: Active fault condition
5. Last: - Previous fault condition

4.5.7. Program Display

1. Program: - Program name of selected program
2. Steps: Total number of steps in the program
3. Step description: Displays current step, function of the step (Time, On, Off) and progress in completing the step. This shows number of minutes in the current step
4. Next: Next step function

4.5.8. STP Display

1. Ta: Measured ambient temperature
2. Pa: Measured barometric pressure
3. Fa: Ambient conditions flow rate
4. Va: Ambient conditions sample volume
5. Ts: Standard conditions temperature
6. Ps: Standard barometric pressure
7. Fs: Standard conditions flow rate
8. Vs: Standard conditions sample volume

4.6. Sensor Calibration

The GilAir Plus pump has a built in automatic sensor calibration feature which allows the pump to maintain an accurate stable flow rate by intermittently calibrating the pump's flow sensor to establish the zero flow point. This routine occurs before starting, when the internal temperature of the pump changes more than 3 degrees C, or whenever one hour has elapsed since the last sensor calibration. The calibration takes approximately 15 seconds during which time the pump stops running. Sample time and volume are not counted during the sensor calibration routine as the pump automatically accommodates this brief stoppage in the accumulation of sample results. The message "Sensor Calibration" displays during this process.

4.7. Run Mode

4.7.1. Run Mode Descriptions

Selecting **Run** begins a sampling event. The event data is cleared. Before selecting Run, the control mode, constant flow or constant pressure should have the proper setting and the flow rate or control pressure should be at the desired set point. If **Manual Run Mode** is active, the pump starts in the selected control mode at the set point. If the **Timed Run Mode** is selected, the pump waits for the start time and then starts and runs for the specified duration. If the **Program Run Mode** is selected, the program starts and controls sampling until the program completes. When running in the high flow range, a "Connect Media" message will pop up on the screen if the pump does not see the expected back pressure caused by a sample train.

The **Stop** menu can be accessed by the  button during Run to pause or stop the event. If **Pause** is activated the pump stops with flow, timing, and total volume accurately maintained. The **Stop** menu item terminates the sample event. The keypad can be locked to avoid event modification.

4.7.2. Locking the Keypad

To lock the keypad:

Press and hold down simultaneously the  and  buttons. A lock symbol will appear in the top status line. Release the buttons and the pump will be locked. After locking the pump it will only respond using the  and  buttons to access the Idle, Fault, Program and Standard Conditions status screens.

4.7.3. Unlocking the Keypad

To unlock the keypad:

Press and hold down simultaneously the **+** and **-** buttons. The lock symbol will disappear from the top middle portion of the screen. Release the buttons, and the pump will be unlocked.

IMPORTANT! – The above instructions apply *only* if no Password (Section 5.11.) has been selected for the pump (the Password is set at the factory default value of 0, disabled). If a Password has been previously selected, after the user has performed the instructions above, the keypad will prompt the user to enter the Password, then press and release the **✓** button before the pump will unlock itself.

4.8. Flow Set (cc/min)

Flow Set allows the flow rate at which the pump will operate in constant flow control mode to be set. The allowable range is 20 cc/min to 5000 cc/min. Flow rates between 20 cc/min and 449 cc/min require the flow control valve to be set for low flow operation (**Lo** indicator is displayed in the bottom status line). Above 449 cc/min the flow control valve must be set for high flow operation (**Hi** indicator is displayed in the bottom status line). See the illustrations below for changing the position of the flow control valve.

4.8.1. Setting the Flow Rate Range

The flow rate range is set by using the hex key (provided with pump, standard 2 mm or 5/64 inch size) to change the position of the flow control valve. The following illustration shows the flow control valve in the high flow position (450 to 5000 cc/min), and the word **Hi** is indicated in the bottom status line. (See just above the users thumb).



The illustration below shows the flow range set in the low flow position (20 to 449 cc/min), as indicated by the word **Lo** at the bottom of the display screen.



The illustration below shows the flow control valve in between the **Hi** and **Lo** flow positions, where the display shows **Err**, indicating an error. The pump will not operate unless the flow control valve is in the correct **Hi** or **Lo** position for the corresponding flow rate setting.



4.8.2. Setting the Flow Rate

NOTE: This operation is required only if you are *changing* the pump flow rate.

If you are using the previously set flow rate, you will only need to verify it using a calibrated reference flow meter.

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to Flow Set.
2. Use the  and  buttons to set the desired rate of flow.
3. Press the  button to accept the flow rate. Flow set is now complete.

4.9. Field Calibrate

4.9.1. Field Calibration

It is recommended Industrial Hygiene practice to conduct a field calibration prior to and after each field sample. This practice is commonly known as the Pre-Cal and Post-Cal. In this procedure, the sample flow rate is set and verified during the Pre-Cal using an external air flow calibrator, preferably one that is traceable to National Institute for Standards and Technology (NIST). The flow is again verified at Post-Cal using the same air flow calibration device. Procedures for field calibrations are found in the OSHA Technical Manual and in the NIOSH Manual of Analytical Methods (NMAM). Following these published guidelines for field sampling is highly recommended.

The GilAir Plus has a Pre-Cal/Post-Cal mode (See Section 5.3.) accessible through the Set-up menu. This mode will record a Pre-Cal/Post-Cal value for each sample accessible through the PC interface and upon pump data review. Pre-Cal and Post-Cal may be conducted as separate events through the sample run mode, this method will not include calibration data as events preceding and following the sample event in the datalog.

4.9.2. Calibration Option (SmartCalSM)

An optional automatic user calibration kit is available for the GilAir Plus. The automatic field calibration, known as SmartCal, provides communication between the pump and the calibrator allowing the pump to self-adjust its flow rate. Although this capability is enabled on all pump models, a communications capable dock is required to access the SmartCal feature. A communications capable dock is supplied as a standard accessory only with the datalog (DL) and STP models. A unique communications cable is required for each specific calibration device. When using this feature, and the Pre/Post Cal selection, calibration data is recorded in the transferable event datalog. This option is available for the calibrators listed on the following table:

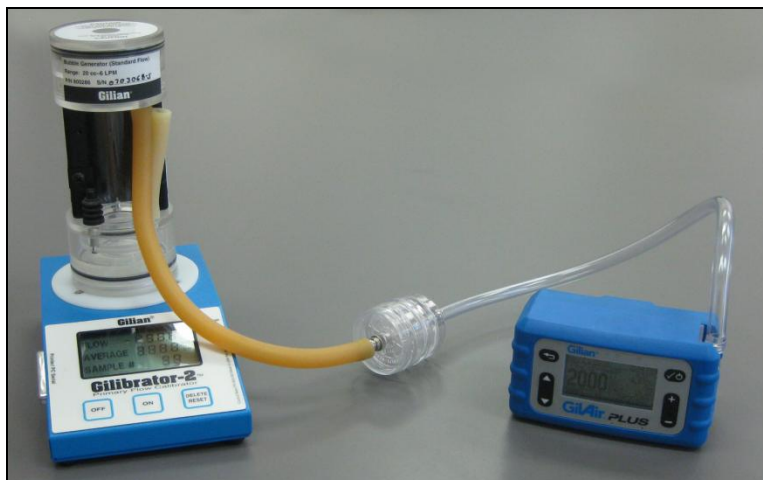
Air Flow Calibrator	Communications Cable	Cable Part Number
Gilian Gilibrator-2	Calibrator cable (Gilibrator), for automatic calibration option	780-0015-01-R
Gilian Challenger®	Calibrator cable (Challenger), for automatic calibration option	780-0015-02-R
TSI™ (Model 4140)	Calibrator cable (TSI), for automatic calibration option	780-0015-03-R
BIOS Defender™	Calibrator cable (BIOS Defender), for automatic calibration option	780-0015-04-R

Note: TSI is a trademark of TSI, Incorporated. Challenger (available through Sensidyne) is a registered trademark of BGI, Incorporated. Defender is a trademark of BIOS, Incorporated.

See Section 5.6. for **Automatic Calibration** instructions.

4.9.3. Displayed Flow Calibration

Calibrate allows the flow rate of the pump to be verified or adjusted to the desired operating point. The calibration will be stored and used in subsequent operation until the pump is recalibrated. Calibration is performed with an air flow calibrator and pressure panel or representative sampling media. The pressure panel should be connected to the inlet of the pump and the reference meter connected to the pressure panel inlet. Calibration always takes place at the selected flow rate.



Calibrating the Displayed Flow Rate:

1. Set up a flow reference instrument (e.g., Gilibrator-2 or Challenger) following the manufacturer's instructions and recommendations.
2. Choose a sample media of similar back pressure to that intended for use in the field.
3. Attach ¼ inch ID tubing from the pump to the sample media, and then from the sample media to the reference flow meter.
4. **NOTE:** To exit Calibrate without changing any values, simply press the  button.
5. Go to the Main Menu screen. Using the  and  buttons, move the cursor  to Calibrate. Press and release the  button.
6. The pump begins to run. Measure the flow rate using the reference flow meter. Use the  and  buttons to adjust the flow rate shown on the pump display to match the actual flow rate measured on the reference meter. Press and release the  button. Using the information of the actual flow rate, the pump will adjust to run at the calibration point. The displayed flow rate will return to the calibration flow rate.
7. Measure the flow rate again using the reference flow meter. If necessary, adjust the flow rate shown on the pump display to match the actual flow rate measured on the reference meter. Press and release the  button.
8. Repeat the adjustment above until the measured flow rate is within 5% of the desired rate of flow. Calibration is now complete. Press the  button to exit calibration.

Note on User Calibration

The above display calibration procedure makes internal pump adjustments and improves the accuracy of the flow display. It does not replace field calibration as described by OSHA and NIOSH. A flow verification using the Gilibrator or other calibrator and the exact field sampling train should be conducted before and after each field sample. Procedures for field calibration may be referenced in the *NIOSH Manual of Analytical Methods* at www.cdc.gov/niosh or in the *OSHA Technical Manual* at www.osha.gov (See Section 4.9.1.).

4.10. Fault Condition cause and displays

When the GilAir Plus encounters operating conditions outside of the pump's operating specifications it will respond by activating the pump's fault system. The fault system will indicate a fault condition by changing the flashing green LED to a flashing red LED. After five seconds in a fault condition, the pump will display the Fault screen detailing the cause of the fault. If the fault condition persists continuously for 30 seconds, the pump will shut the sample off to prevent collection of bad data. There will be no LED illumination during the pause. If fault retry is enabled (See Section 5.4.), the GilAir Plus will attempt an automatic re-start after three minutes. During that time if the fault cause was corrected (e.g., user sat in a chair and pinched the connection tubing shut momentarily) the pump will resume sampling. When the condition remains after ten restart attempts, the pump will end the event. If fault retry is disabled, the sampling event will end when the pump stops after 30 seconds of operation in fault and no retry will be attempted.

Several conditions can initiate a fault condition, as described below:

Flow Fault: If the pump is running in the **Constant Flow Control** mode, and the flow rate cannot be maintained within the constant flow rate specification, the pump will go into a Flow Fault, a fault caused by flow outside the specifications of the pump.

Over Pressure Fault: If the pump is running in the **Constant Flow Control** mode and the sample media has increased in flow resistance (i.e., back pressure) due to the accumulated sample or a blockage in the sample train, the pump will go into a fault condition after reaching its specified back pressure maximum.

Pressure Fault: If the pump is running in the **Constant Pressure** mode and the sample back pressure cannot be maintained within +/- 10% of the set point pressure the pump will go into a Pressure Fault condition. This is usually caused by insufficient sample resistance, which causes flow rates beyond the operating range of the pump.

Low Battery Fault: If the battery voltage has dropped below a minimum level, the pump will go into a fault condition due to a low battery. Restart attempts will not be made for this fault.

Valve Fault: If the High Flow/Low Flow mode control valve is not at the correct setting, either positioned between the high and low settings or set at the incorrect setting for the flow rate selected the pump will go into a Valve Fault. The correct High Flow/Low Flow setting must be made to resolve the fault. (Section 4.8.1.)

Memory Fault: If the memory for the datalog is full a memory fault will occur. The datalog must be emptied before the pump can be restarted. The data memory is large and this fault is unlikely.

SECTION FIVE: Options

5.1. Setup ►

The **Setup** submenu has controls that adjust the basic operating parameters of the pump.

To change the settings for each item in the **Setup** submenu:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen.
2. At the **Setup** submenu screen, use the  and  buttons to move the cursor ► to the submenu item for which you desire to change the setting. Use the  and  buttons to select the setting desired for that item, then press and release the  button. The change to the new setting is now complete.

Note: Many setup menu items are in the submenus.

5.2. Event ID Enable

An event is defined as one sample run. **Event ID Enable** enables or disables the collection of sample identification information that will be stored in the event record to identify the event. When the event is enabled the pump will prompt for an ID each time an event is started. The pump will remember the event number and propose it for modification when the next event ID is required. The ID is available in **Review** and via data communications.

To utilize the **Event ID enable** function:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen.
2. At the **Setup** submenu screen, the cursor ► will now be at Event enable. The default setting for Event enable is disable. Press and release the  or  buttons to change the setting to enable. Press and release the  button.

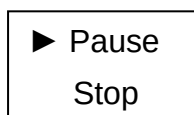
Note: Event ID is enabled when Run is used to start an event. The pump will prompt the user for an ID in Manual, Timed or Program operation modes.

3. Press and release the  button to return to the Main Menu. Use the  button to move the cursor to Run. Press and release the  button.

4. Event ID 1 will now be displayed. Use the  or  buttons to assign the event any unique identification number from 1 to 999,999.

Note: A single press and release of either of these buttons will increment or decrement the ID number by 1. Press and hold down either button to scroll through the full range of numbers. When the ID number is selected, press and release the  button to complete.

5. Upon pressing and releasing the  button, the pump will begin to run at the flow rate selected at Flow Set (see Section 4.8.). To stop the run, press and release the  button again.
6. The following message box will appear:



7. Press and release the  button to move the cursor to Stop. Press and release the  button. The pump stops running.
8. At the Main Menu, use the  button to go to Review. Press and release the  button. The run data for the event is displayed: Flow mode setting; flow rate; volume sampled; start time and date; Pre-Cal flow rate and event ID number.

5.3. Pre/Post-Calibration

Pre/Post-Calibration allows the user to record both a Pre-Cal and Post-Cal rate of flow for a sample. The default setting for this function is **disable**. When enabled, Pre and Post-Cal flow readings will be taken in Manual, Timed and Program operating modes.

To enable the **Pre/Post-Cal** function:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen.
2. At the **Setup** submenu screen, use the  and  buttons to move the cursor ► to **Pre/Post Cal**. Use the  and  buttons to select **enable**. Press and release the  button to complete.

This completes the enable of the Pre/Post Cal option. The description of the use of the option when starting an event is given below.

3. When you are at **Run** in the Main Menu and press the  button to start a sample run, the calibration screen displays the set flow rate for the sample (See Section 4.8.). Using a calibrated flow measurement reference (e.g., Gilibrator-2 or Challenger), measure the pump's flow rate, then adjust the displayed flow rate on the pump, using the  and  buttons, to match the flow rate displayed on the reference flow meter. Press and release the  button.
4. The pump continues to run showing the Constant Flow display screen. Complete the sample run, then press and release the  button. A **Pause/Stop** message box appears. Use the  button to move the cursor ► to **Stop**. Press and release the  button.

Note: If the event ends in Program or Timed operating modes, the Post-Cal will be requested at that time.

5. The pump stops running and the following message appears on the screen:

Ready for Post Cal

Press Enter

Press and release the  button. The pump begins to run again, and the calibration screen is displayed showing the validation flow rate for the sample (See Section 4.6.). Using the calibrated flow measurement reference, measure and enter the pump's flow rate. Press and release the  button.

6. To review the sample go to the Main Menu screen, use the  and  buttons to move the cursor ► to **Review** press and release the  button. The screen shows the sample with the Pre-Cal flow rate at the bottom left portion of the screen (marked **Pre:**). Press and release  button. The second screen of the sample is shown with the Post-Cal flow rate at the bottom left portion of the screen (marked **Post:**).

Note: SmartCal can be used to supply the Pre/Post-Cal data.

5.4. Fault Retry

When **Fault Retry** is enabled, the pump will attempt an auto restart every 3 minutes after it has encountered a condition where it has gone into fault and stopped running. The pump will attempt restart for up to 30 minutes with a total of 10 attempted restarts. The default setting for **Fault Retry** is **enabled**.

To disable **Fault Retry**, so that the pump will not attempt a restart after faulting off:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen.
2. At the **Setup** submenu screen, use the  and  buttons to move the cursor ► to **Fault Retry**. Use the  and  buttons to select **disable**. Press and release the  button to complete.

5.5. Valve Mode

The **Valve Mode** function has 2 settings:

The **continuous** setting allows the pump to check itself continuously during a sample run to detect an incorrect setting of the flow control valve which controls the low and high flow modes of the pump. This is the default setting for the **Valve Mode** function.

The **start/stop** setting is utilized when the pump is used in an environment where it might be exposed to extreme magnetic interference. When **Valve Mode** is in **start/stop**, the pump will only check the positioning of the flow control valve at the beginning and end of a sample run to minimize any adverse effect from magnetic interference.

To put the **Valve Mode** into **start/stop**:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen.
2. At the **Setup** submenu screen, use the  and  buttons to move the cursor ► to **Valve Mode**. Use the  and  buttons to select **start/stop**. Press and release the  button to complete.

5.6. SmartCalSM Automatic Calibration

GilAir Plus has an optional automatic calibration capability called SmartCal. Select an air flow calibrator and connect it to the Dock using the appropriate communications cable. When active; Pre, Post and flow calibration can be performed automatically through the Dock. This option requires the communications version of the Dock, (standard with DL and STP models). (See Section 4.9.2. for a list of calibration devices and communications cables).

The tubing setup is the same as a manual calibration (See Section 4.9.3.). Using the SmartCal feature requires the GilAir Plus pump to be connected to the Dock with the cable to the calibrator plugged in.

After connections are complete, the desired calibrator must be selected in the **Setup** ► **SmartCal** menu item. After the setup is complete, select the flow rate for the specific calibration needed. When **Calibrate** is selected from the main menu, the pump enters SmartCal and displays the type of calibrator and the current flow rate reading. Follow the calibrator instructions below to begin the calibration:

5.6.1. Gilibrator-2

Once the flow rate stabilizes, generate a series of 10 bubbles, making 10 measurements. If a bad bubble occurs, delete the bubble using the reset key on the Gilibrator. When the GilAir Plus receives the tenth bubble within an acceptable range of deviation, the calibration will be calculated and stored. *Calibration Complete* will be displayed. The  button can be used to return to the main menu.



1. Connect the SmartCal cable to the PC Serial connector on the Gilibrator and the Reference Device port on the rear of the Dock.
2. Using the GilAir Plus pump main menu, go to **Setup** ► **SmartCal**. Use the  and  buttons to select **Gilibrator**. Press and release the  button. Go to **Flow set**. Use the  and  buttons to set the flow rate. Press and release the  button. Go to **Calibrate**, then press and release the  button. The pump will begin to run in the SmartCal mode.
3. While running in the SmartCal mode before readings are taken on the Gilibrator the following is displayed:
SmartCal
Gilibrator 0 cc/m
4. Take 10 readings on the Gilibrator. After the 10th reading is taken allow a moment for the pump to adjust its flow rate. Refer to the Gilibrator user manual to locate instructions for taking a reading.

5. The flow rate reading displayed on the pump matches the flow rate reading displayed on the Gilibrator for each of the 10 readings taken. After allowing the pump to adjust the flow rate, the pump stops running when the flow rate displayed on the pump is adjusted to within $\pm 5\%$ of the selected flow rate. *Calibration Complete* is displayed and the pump stops.
6. **Before removing the pump from the Dock**, press and release the  button. The Main Menu screen is displayed. Remove the pump from Dock. Go to **Run**. Press and release the  button. The pump begins to run at the selected flow rate.

5.6.2. Challenger[®]

Verify the desired calibration flow rate is within the Challenger range. Select the appropriate range if necessary. The Challenger reads flow continuously and reports it for display on the GilAir Plus. When flow stability is reached, the calibration will be calculated and stored in the pump. *Calibration Complete* will be displayed. The  button can be used to return to the main menu.

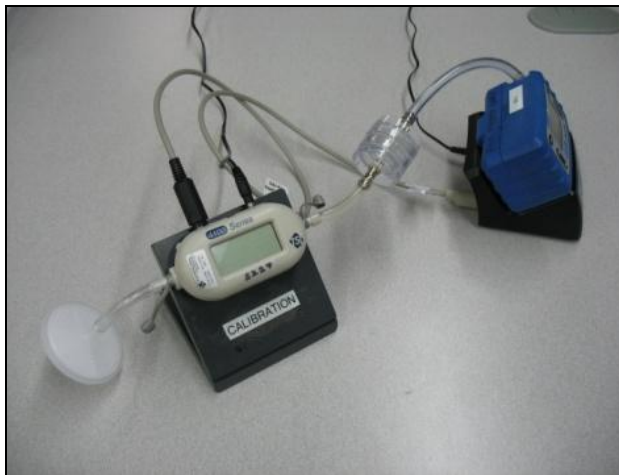


1. Connect the SmartCal cable between the RS232 port on the Challenger and the Reference Device connector on the rear of the Dock.
2. Check that the Challenger is set to the range required for the calibration measurement. Power on the Challenger before connecting the Challenger to the Dock via the cable, and leave the Challenger undisturbed as it self-calibrates. When “Qa” and “Qs” values are displayed on the Challenger connect the hose from the GilAir Plus air sampler to the Challenger and connect the SmartCal cable to the Challenger and Dock.

3. On the GilAir Plus pump, go to **Setup**, then **SmartCal**. Use the **+** and **-** buttons to select **Challenger**. Press and release the **✓** button. Go to **Flow set**. Use the **+** and **-** buttons to set the desired flow rate. Press and release the **✓** button. Go to **Calibrate**, then press and release the **✓** button. The pump will begin to run in the SmartCal mode.
4. While running in the SmartCal mode, the Challenger continuously displays the flow rate, and the following is displayed on the pump:
SmartCal
Challenger (flow rate) cc/m
5. The GilAir Plus will run for a few minutes adjusting its flow rate to the calibration value using the readings from the Challenger. *Calibration Complete* will be displayed and the pump will stop.
6. **Before removing the pump from the Dock**, press and release the **↩** button. Remove the pump from the Dock and disconnect the cable from the Challenger then go to **Run** in the Main Menu, and press and release the **✓** button.
7. The pump begins to run at the set flow rate.

5.6.3. TSI™ Model 4140

The TSI Model 4140 reads flow continuously and reports it for display on the GilAir Plus. When flow stability is reached, the calibration will be calculated and stored. *Calibration Complete* will be displayed. The **↩** button can be used to return to the main menu.



1. Connect the SmartCal cable between the interface port on the TSI Model 4140 and the Reference Device port on the rear of the Dock.

2. On the GilAir Plus pump, go to **Setup**, then SmartCal. Use the **+** and **-** buttons to select **TSI**. Press and release the **✓** button. Go to **Flow set**. Use the **+** and **-** buttons to set the desired flow rate. Press and release the **✓** button. Go to **Calibrate**, then press and release the **✓** button. The pump will begin to run in the SmartCal mode.
3. While running in the SmartCal mode, the TSI 4140 displays the flow rate, and the following is displayed on the pump:
SmartCal
TSI (flow rate) cc/m
4. When flow stability is reached, the calibration will be calculated and stored. *Calibration Complete* will be displayed.
5. **Before removing the pump from the Dock**, press and release the **←** button. The Main Menu screen is displayed. Remove the pump from Dock. Go to **Run**. Press and release the **✓** button. The pump begins to run at the set flow rate.

5.6.4. Bios Defender™ 510

The Defender has a limited flow range, verify that the desired calibration flow rate is within range. The Defender is set to cycle continuously and read flow. When adequate samples are collected to guarantee flow stability, the calibration will be calculated and stored. *Calibration complete* will be displayed. The **←** button can be used to return to the main menu.



1. Setup the GilAir Plus and Bios Defender with the SmartCal cable connected to the Data port on the Defender and the Reference Device connector on the rear of the Dock as shown in the photo.

2. Set up the BIOS Defender to measure 10 samples continuously. On the GilAir Plus pump, go to **Setup**, then **SmartCal**. Use the  and  buttons to select **BIOS Dfndr**. Press and release the  button. Go to **Flow set**. Use the  and  buttons to set the desired flow rate. Press and release the  button. Go to **Calibrate**, then press and release the  button. The pump will begin to run in the SmartCal mode.
3. While running in the SmartCal mode, the BIOS Defender displays the flow rate for each sample taken, and the following is displayed on the pump:

SmartCal
BIOS Dfndr (flow rate) cc/m

4. When adequate samples are collected to guarantee flow stability, the calibration will be calculated and stored. *Calibration complete* will be displayed.
5. **Before removing the pump from the Dock**, press and release the  button. Then remove the pump from the Dock, go to **Run** in the Main Menu, and press and release the  button. The pump begins to run at the set flow rate.

5.7. Clear Datalog

Clear Datalog erases all datalog entries. When the datalog is full, any data saved in subsequent events will be lost. The datalog should be cleared after the data is extracted to allow space for new events. **The Datalog has the capacity to store up to 16 events.** The number of slots consumed in the datalog is shown on the Idle screen. Starting an event without space in the datalog will cause a warning screen that must be acknowledged in order to perform an event that will not be stored in the datalog.

To clear the datalog:

IMPORTANT! Verify that all desired data has been archived prior to clearing!

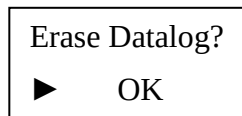
1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen.
2. At the **Setup** submenu screen, use the  and  buttons to move the cursor ► to **Clear Datalog**. Press and release the  button. The following message box will be displayed:

Erase Datalog?

► Cancel

Note: To cancel out of **Clear Datalog**, press and release the  button at this time.

3. Press and release the  or  button. The message box now displays:



4. Press and release the  button. The Datalog is now erased.

5.8. Run Options ►

5.8.1. Standard Temperature (STP Models only)

Users may set the temperature that will be used in the calculation of the standard flow and volumes when using a STP Model pump. The GilAir Plus factory default is 25°C (77°F) as the standard temperature.

To set the Standard Temperature:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen.
2. At the **Setup** submenu screen, use the  and  buttons to move the cursor ► to **Run Options** ►. Press and release the  button.
3. The cursor ► will now be at **Std Temp (° C or °F)**. The current setting for **Std Temp (° C or °F)** will be displayed at the right side of the screen. Use the  and  buttons to adjust the setting to the temperature desired.

Note: A single press and release of either of these buttons will increment or decrement the temperature setting by 1 degree. Press and *hold down* either button to scroll through the full range of numbers. When the desired temperature setting is selected, press and release the  button to complete.

5.8.2. Standard Pressure (mmHg) (STP models only)

Users may set the pressure that will be used in the calculation of the standard flow and volumes when using a STP model pump. The GilAir Plus factory default is 760 mmHg as the standard pressure.

To set the Standard Pressure:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen.
2. At the **Setup** submenu screen, use the  and  buttons to move the cursor ► to **Run Options** ►. Press and release the  button.
3. The cursor ► will now be at **Std Temp (° C)**. Press and release the  button to move the cursor ► to **Std P (mmHg)**. The current setting for **Std P (mmHg)** will be displayed at the right side of the screen. Use the  and  buttons to adjust the setting to the pressure desired. **Note:** A single press and release of either of these buttons will increment or decrement the pressure setting by 1. Press and *hold down* either button to scroll through the full range of numbers. When the desired pressure setting is selected, press and release the  button to complete.

Note: mmHg is the only pressure unit used for standard pressure.

5.9. Display Options ►

These parameters control the appearance and format of displayed pump data.

5.9.1. Language

Users may select the display language; languages available include: English, Spanish, French, Dutch, German, Italian, and Portuguese. English is the default.

To select the display language:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen.
2. At the **Setup** submenu screen, use the  and  buttons to move the cursor ► to **Display Options** ►. Press and release the  button.
3. The cursor ► will now be at **Language**. Press and release either the  or  button. Each press and release allows you to select from the following choices of languages: English, Spanish, French, Dutch, German, Italian, and Portuguese. When the desired language is selected, press and release the  button to complete.

5.9.2. Temperature Units

Set the displayed temperature unit to **Celsius** or **Fahrenheit**.

To set the units in which temperature is displayed:

1. From the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen.
2. At the **Setup** submenu screen, use the  and  buttons to move the cursor ► to **Display Options** ►. Press and release the  button.
3. Press and release the  button to move the cursor ► to **Temperature Units**. Press and release either the  or  button. Each press and release allows you to select either °C or °F. When the desired unit is selected, press and release the  button to complete.

5.9.3. Pressure Units

Set the displayed unit of pressure for backpressure to **”H₂O**, **mmHg** or **KPa**.

To set the units in which pressure is displayed:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen.
2. At the **Setup** submenu screen, use the  and  buttons to move the cursor ► to **Display Options** ►. Press and release the  button.
3. Press and release the  button to move the cursor ► to **Pressure Units**. Press and release either the  or  button. Each press and release allows you to select **”H₂O**, **mmHg** or **KPa**. When the desired unit is selected, press and release the  button to complete. In display screens, inches of water is abbreviated ", mmHg is abbreviated Hg and KPa is abbreviated KP.

5.10. Clock Set ►

Set the date, time, and hour format.

5.10.1. Clock

To set the time of day:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen.
2. At the **Setup** submenu screen, use the  and  buttons to move the cursor ► to **Clock Set** ►. Press and release the  button.

3. The cursor ► will now be at **Clock**. Use the  and  buttons to set the hour. When the desired hour is selected, press and release the  button to complete the hour portion of the **Clock Set** function. **Note:** The hour must be selected per the standard 24-hour format (i.e., 1:00PM = 13:00).
4. After the  button has been pressed and released in step 3 above, the present value for minutes will be displayed. Use the  and  buttons to set the minutes. When the desired value for minutes is selected, press and release the  button to complete the minute portion of the **Clock Set** function. The selected time will now be displayed at the top right portion of the display screen.

5.10.2. Date

To set the Date:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen. At the **Setup** submenu screen, use the  and  buttons to move the cursor ► to **Clock Set** ►. Press and release the  button.
2. Press and release the  button to move the cursor ► to **Date**. Use the  and  buttons to set the year. When the desired year is selected, press and release the  button to complete the year portion of the **Date** function.
3. After the  button has been pressed and released in step 2 above, the *month* will be displayed. Use the  and  buttons to set the month. When the month is selected, press and release the  button to complete the month portion of the **Date** function.
4. After the  button has been pressed and released in step 3 above, the *day* will be displayed. Use the  and  buttons to set the day. When the day is selected, press and release the  button to complete setting of the **Date** function. The selected month, day and year will now be displayed in the top left portion of the screen.

5.10.3. Time Format

Choose between 12-hour and 24-hour clock display.

Note: Some displays always default to the 24-hour time display due to space limitations.

To select the displayed **Time Format**:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen.
2. At the **Setup** submenu screen, use the  and  buttons to move the cursor ► to **Clock Set** ►. Press and release the  button.
3. Press and release the  button to move the cursor ► to **Time format**. Use the  and  buttons to select either the **12h** or the **24h** format. When the desired time format is selected, press and release the  button to complete. The time will now be displayed in the selected format in the top status line.

5.10.4. Date Format

Select between MM/DD/YY and DD/MM/YY date formats.

To set the Date Format:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen.
2. At the **Setup** submenu screen, use the  and  buttons to move the cursor ► to **Clock Set** ►. Press and release the  button.
3. Press and release the  button to move the cursor ► to **Date format**. Use the  and  buttons to select either the **mm/dd/yy** or the **dd/mm/yy** format. When the desired date format is selected, press and release the  button to complete. The date will now be displayed in the selected format in the top left portion of the screen.

5.11. Password

The GilAir Plus offers a password protection feature that can be set but is not required. The password for unlocking the keypad can be set in the menu item. A password may consist of 1-4 numerals. If the password is set to 0, the password will not be required and the keypad will unlock without password entry.

Change password to 0 to remove password.

To set the password:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Setup** ►. Press and release the  button. The pump will now display the **Setup** submenu screen.

2. At the **Setup** submenu screen, use the  and  buttons to move the cursor ► to **Password**. Use the  and  buttons to assign any unique number from 0 to 9999 as a password. Press and *hold down* either button to scroll through the full range of numbers. When the *password* is selected, press and release the  button to complete.

5.12. Control Mode

CF (Constant Flow), CPH (Constant Pressure High), CPL (Constant Pressure Low)

The **Control Mode** determines which control parameter the GilAir Plus holds constant during the sample. The two control mode options are holding the flow constant or holding the pressure constant. Both control modes will run in high flow (450-5000 cc/min) and low flow (1-449 cc/min) mode. **Constant Flow** control should be chosen for single samples, and **Constant Pressure** control should be chosen when splitting the flow into multiple samples. In the **Constant Flow** mode, both the flow rate and the back pressure are displayed. In the **Constant Pressure** mode only the back pressure is displayed. Constant pressure control requires sample media holders with needle valves to control the flow rate. In the constant pressure mode, the flow rate is read at the external calibrator only.

Constant Flow Control and **Constant Pressure Control** are fundamentally different modes of operation. A description of the two modes follows.

Constant Flow Control holds the flow rate constant throughout the sample duration, even in the presence of variations in the back pressure of the sample media often caused by back pressure buildup on the filter.

Constant Pressure Control maintains a constant pressure level at the inlet of the pump, which provides a stable vacuum for drawing the sample. A needle valve is required in the sample media holder to set the flow rate. If the sample media resistance is constant, the flow through the media will also remain constant.

This mode allows the sample to be split, and it is also called the **Multi-Flow Mode**. This mode requires a sample media holder that incorporates a needle valve in each split segment.

Example; a dual holder will require two needle valves to set the two flow rates independently. The constant pressure source allows the flow to be split, because adjusting one side does not change the flow rate through the other side, since the collective force pulling the two samples remains constant.

Constant pressure mode maintains constant flow in the sample only when the flow resistance of the sample path remains constant during sampling. If the flow resistance varies the flow rate will vary. With the default pressure set point of 18 inches of water, if the pressure varies by 0.9 inches of water, the flow will change by about 5%. Care should be used in selecting the applications that are used with constant pressure control and the behavior of the sample resistance under field conditions should be verified.

To select the **Control Mode**:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Control Mode**. Press and release the  button.
2. Use the  and  buttons to select either the **CF** (Constant Flow [20-5000 cc/min]), **CPL** (Constant Pressure Low [1-449 cc/min]) or the **CPH** (Constant Pressure High [450-5000 cc/min]) control mode. When the desired control mode is selected, press and release the  button to complete. The selected control mode will now be displayed in the bottom status line.

Note: The flow control valve must be set to the **Lo** position for running from 1-449 cc/min, and it must be set in the **Hi** position for running above 450 and up to 5000 cc/min for both the constant flow and constant pressure control modes.

5.13. Run Mode Manual, Timed, Program Name

The **Run Mode** selection determines how the timing of the sample is managed. In **Manual mode**, the pump is switched on and off with the keypad. Any run time is possible within the limits of the pump capacity. In **Timed mode** the start time and duration of sampling is specified in the **Run Setup** submenu. In **Program mode**, the flow rate and a series of clock times, on-periods and off-periods can be programmed. The program is entered in **Program edit** under **Run Setup**. Each Program has a name and the names are shown in this menu item selection to allow the desired run parameters to be set.

Note: The Program Mode is offered only in the Datalog and STP models. The Timed mode is a feature of all three models.

To set the **Run Mode**:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Mode**. Press and release the  button.
2. Use the  and  buttons to select either the **Manual** mode or the **Timed** mode on a program. When the desired run mode is selected, press and release the  button to complete.

Note: If one or more programs have been entered and saved, then those programs will be listed as available **Run Mode** selections *in addition to* the **Manual** and **Timed** functions and can be selected the same way. If no programs have been entered and saved, then no program names will be shown.

Note: When the pump is first powered on, it will remember the last set Run Mode, Manual, Timed or selected Program. If a different mode is desired, it must be set before running an event.

5.14. Run Setup ►

Go to **Run Setup** to set the parameters for constant pressure mode operation, timed and programmed runs.

To enter Run Setup:

At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Setup** ►. Press and release the  button.

5.14.1. Timed Start

Select a specific clock time as the beginning time of a sample event. At the selected time, the pump will sample for the timed duration as specified below.

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Setup** ►. Press and release the  button.
2. The cursor ► will now be at **Timed start**. On the display screen, to the right of **Timed start**, a clock time is displayed in hours and minutes. The hour portion is underlined. Use the  and  buttons to set the hour. When the hour is selected, press and release the  button. **Note:** The hour must be selected per the standard 24 hour format (i.e., 1:00PM = 13:00).
3. The minute portion of the clock time will now be underlined. Use the  and  buttons to set the minutes. When the minutes are selected, press and release the  button. The setting of the start time for the timed run is now complete.

Note: When a timed run is started and the time is set to 00:00, the event will start immediately.

5.15. Timed Duration

Timed Duration specifies the time in minutes for the pump to run a timed event.

To set the sampling duration of the timed run:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Setup** ►. Press and release the  button.

2. Use the  and  buttons to move the cursor ► to Timed Duration. Use the  and  buttons to set the time for the duration of the run in minutes. When the run time in minutes is selected, press and release the  button. The setting of the run time for the timed run is now complete.

Note: See Section 6.1. and Appendix B for advanced programming the DL and STP models.

5.16. Pressure Set ("H₂O or mmHg or KPa)

Pressure set is used to set the pressure that will be maintained when the pump is running in the constant pressure (**CPL or CPH**) control mode (See Section 5.12.). Available pressure set points are between 5 inches of water to 40 inches of water. If the pressure is set outside the performance capability of the pump, a fault will occur when **Run** is selected.

Note: Factory default is 18 inches of water.

To set the pressure that will be maintained in the Constant Pressure Mode:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Setup ►**. Press and release the  button.
2. Use the  and  buttons to move the cursor ► to **Press set ("H₂O, mmHg or KPa)** (See Section 5.9.3. to select units). Use the  and  buttons to set the pressure. When the pressure is selected, press and release the  button to complete.

SECTION SIX: Programming

6.1. Program ►

User programs specify a timed series of actions that control the sampling protocol employed by the pump. The pump can store up to 16 programs, each of which can be created, edited and executed. A program allows the specification of flow rate, control mode and the timed steps that determine when the pump runs.

Note: This advanced level of programming is available only in the Datalog and STP models.

IMPORTANT! – FOR AN ENTERED PROGRAM TO BE ABLE TO RUN:

The entered program must be *assigned a program name* (Section 6.4.), it must be *saved* (Section 6.10.), the Program Enable function must be set to *enable* (Section 6.2.), and Run Mode must be set to *that program name* (Section 5.13.).

Note: An example program detailing the procedures for entering and editing can be found in Appendix B.

Note: Users can create, edit and load programs the PC Application.

To navigate to the **Program** function:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Setup ►**. Press and release the  button.
2. Use the  and  buttons to move the cursor ► to **Program ►**. Press and release the  button.

6.2. Enabling the Program Mode

To **enable** the pump to run a program:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Setup ►**. Press and release the  button.
2. Use the  and  buttons to move the cursor ► to **Program ►**. Press and release the  button.

3. The cursor ► will now be at **Enable**. The default setting displayed for the **Enable** function is **disable**. Use the  and  buttons to select **enable**. When **enable** is selected, press and release the  button to complete.

6.3. Program Edit ►

This submenu allows the creation and editing of up to 16 programs. Each program is stored in nonvolatile flash memory and can be selected using the **Run mode** item on the main menu (See Section 5.13.).

IMPORTANT! Programs that are created or modified must be saved for the changes to be retained.

To navigate to the **Program Edit** function:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Setup ►**. Press and release the  button.
2. Use the  and  buttons to move the cursor ► to **Program ►**. Press and release the  button.
3. Use the  and  buttons to move the cursor ► to **Program Edit ►**. Press and release the  button.

6.4. Program Name

The program name is displayed when the **Run mode** is selected after a program has been entered and saved. When the program is entered on the pump, a name is assigned, i.e. – **PROG 1** through **PROG 16**. **Note:** Using the *data communication option and a PC*, alphanumeric program names may also be assigned.

To select a **Program Name** for the program:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Setup ►**. Press and release the  button.
2. Use the  and  buttons to move the cursor ► to **Program ►**. Press and release the  button.
3. Use the  and  buttons to move the cursor ► to **Program Edit ►**. Press and release the  button.

4. The cursor is now at **Prog Name**. At the right of **Prog Name**, **PROG 1** is displayed. Use the  and  buttons to select the **Prog Name**, then press and release the  button to complete.

6.5. Control Mode

Use to select either the **Constant Flow** or **Constant Pressure** control mode for a program. The **Set point** function is used to set the flow rate or pressure depending on the setting of this parameter. The **Constant Pressure Mode** must specify either High flow mode (CPH) or Low Flow mode (CPL).

To select the **Control Mode** for a program:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Setup ►**. Press and release the  button.
2. Use the  and  buttons to move the cursor ► to **Program ►**. Press and release the  button.
3. Use the  and  buttons to move the cursor ► to **Program Edit ►**. Press and release the  button.
4. Use the  and  buttons to move the cursor ► to **Control Mode**. Use the  and  buttons to select the control mode in which you want the pump to run during the execution of the program, then press and release the  button to complete.

6.6. Set the Flow Rate

The flow rate set point for constant flow is selected in the **Program Edit** submenu. The same limits and restrictions apply in **Program Mode** as in **Manual Mode**.

To program the **Setpoint** flow rate for a program:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Setup ►**. Press and release the  button.
2. Use the  and  buttons to move the cursor ► to **Program ►**. Press and release the  button.
3. Use the  and  buttons to move the cursor ► to **Program Edit ►**. Press and release the  button.

4. Use the  and  buttons to move the cursor ► to **Setpoint**. Use the  and  buttons to select the set point flow rate at which you want the pump to run during the execution of the program, then press and release the  button to complete.

6.7. Program Steps

Each program step is numbered and executed sequentially. To enter or edit a program select the desired program step and edit it to perform the desired function.

To enter or edit a particular program step:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Setup** ►. Press and release the  button.
2. Use the  and  buttons to move the cursor ► to **Program** ►. Press and release the  button.
3. Use the  and  buttons to move the cursor ► to **Program Edit** ►. Press and release the  button.
4. Use the  and  buttons to move the cursor ► to **Prog Step**. Use the  and  buttons to select the program step you wish to enter or edit, then press and release the  button to complete.

6.8. Function

Each program step performs a specific function. The available functions are:

- **Time** – waits for a specified time to arrive before the next program step is executed.
- **On Interval** – turns on the pump for the specified number of minutes.
- **Off Interval** – specifies a time during which the pump does not sample.
- **Cycle** – loops back and repeats the entire program for a specified number of times.

The cycle function is used mainly for intermittent sampling (e.g., 5 minutes on and 55 minutes off for 72 hours).

To set the **Function** for a particular program step:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Setup** ►. Press and release the  button.
2. Use the  and  buttons to move the cursor ► to **Program** ►. Press and release the  button.

3. Use the  and  buttons to move the cursor ► to **Program Edit** ►. Press and release the  button.
4. Use the  and  buttons to move the cursor ► to **Function**. Use the  and  buttons to select the program function you want the pump to perform at a particular program step, then press and release the  button to complete.

IMPORTANT! – Verify that you are at the correct program step (and change it if necessary per Section 6.7) before selecting or changing the function to be performed at that program step.

6.9. Function Value

Specifies the parameter associated with the function selected to be performed at a particular program step. For **Time**, the parameter is a time of day; for **On Interval** and **Off Interval**, the parameter is time in minutes; and for **Cycle**, the parameter is number of cycles.

IMPORTANT! – Verify that you are at the correct program step (and change it if necessary per Section 6.7) and that the correct program function (Section 6.8.) has been entered for that step before selecting or changing the function value to be performed at that program step.

To set the **Function Value**:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Setup** ►. Press and release the  button.
2. Use the  and  buttons to move the cursor ► to **Program** ►. Press and release the  button.
3. Use the  and  buttons to move the cursor ► to **Program Edit** ►. Press and release the  button.
4. After verifying that you have entered the correct program step (Section 6.7.) and the correct program function (Section 6.8.) that you wish to enter or change the function value, use the  and  buttons to move the cursor ► to **Function Value**. If the program step you are entering/changing has the **Time** function assigned, then use the  and  buttons to select first the hour (in 24-hour format), press and release the  button, then use the  and  buttons to select the minute, then press and release the  button to complete.

5. If the program step you are entering/changing has the **On Interval** or **Off Interval** function assigned, use the  and  buttons to select the number of minutes for the duration of that interval, then press and release the  button to complete.
6. If the program step you are entering/changing has the **Cycle** function assigned, use the  and  buttons to specify the total number of cycles for the number of times you want the pump to loop back and repeat the entire program, then press and release the  button to complete.

6.10. Save a Program

Save is used to store the edited program to the nonvolatile memory.

IMPORTANT! All changes will be lost unless the program is saved!

To **Save** a program after it has been entered or edited:

Use the  and  buttons to move the cursor ► to **Save**. To the right of **Save**, you will see the message **Changed** displayed. Press and release the  button. The **Changed** message disappears from the screen. The program has now been saved to memory.

6.11. Review Events

Review allows the pump to display the results of the last 16 sample events. When the item is selected, a display appears that shows the sample number (1 is the most recently stored sample, 16 is the oldest). Each stored event can be displayed by using the  and  buttons.

The display for each event includes control mode, run mode, flow rate or pressure, total sampled volume, run time, pre and post calibrations, and start/stop times.

To **Review** the stored sample events:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Review**. Press and release the  button.
2. Use the  and  buttons to review each sample event. There are two pages of data for each event and the  and  buttons switch between the pages. Press and release the  button to exit **Review** and return to the Main Menu.

SECTION SEVEN: PC Interface

7.1. PC Interface

The GilAir Plus Datalog and STP models have the capability of PC interface through the charging dock. This requires a special communications version dock that is supplied in kits with the Datalog and STP pump models. The basic model of the GilAir Plus does not have this capability, nor will the basic Dock supplied with the Basic model pump support this option.

The PC interface enables the following functions:

- Download of event data, including summary and minute by minute datalogs from the pump to a data base
- Creating of a field report for each sample
- Editing and transfer of settings to the pump
- Export of event historical data in CSV file format for use in a PC spreadsheet application such as Excel™

Connecting the communications dock to a PC requires USB cable P/N 811-0907-01-R. Each docked pump is recognized by its serial number. A new pump must be registered before it will be recognized. The PC interface program format is summarized in the table below. A separate manual, PN360-0143-01, describes the operation of the PC application in detail.

Main Tabs	Sub Tabs	Function
Data Retrieval and Reports	Review Data from Logs	Download Sample Logs & Produce Field Reports and Graphs
Pump Manager	Manage Pump Information & Manage Pump Configuration	Manage pump setup and parameters
Program Manager	Manage Programming & Transfer Programs to Pump(s)	Create, Edit, Duplicate or Delete Sampling Programs; Transfer Programs to Pump(s)

Note: Please see the PC Application User Manual for additional information.

SECTION EIGHT: Maintenance Menu

8.1. Maintenance ►

The **Maintenance** submenu has controls that are occasionally used for clearing the GilAir Plus pump memories and calibrating transducers.

To change the settings for each item in the **Maintenance** submenu:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Maintenance** ►. Press and release the  button. The pump will now display the **Maintenance** submenu screen.
2. At the **Maintenance** submenu screen, use the  and  buttons to move the cursor ► to the desired submenu item. Each of the submenu items have additional menus or actions beneath them and are selected by pressing and releasing the  button.

8.2. Global Reset

Global Reset resets all user controlled parameters to the factory ship values.

Default factory values:

Parameter	Default Value
Flow Rate	2000 cc/min
Control Mode	CF
Run Mode	manual
Point calibrations	nominal
Pressure set point	18 inches H2O
Timed start	00:00 am
Timed duration	01 minutes
Event enable	disabled
Events in Datalog	retains current data
Standard Temperature	25 degrees C
Standard Pressure	760 mmHg
Language display	English
Temperature Units	degrees C
Pressure Units	inches H2O
Password	disabled (0)
User Programs	cleared

To perform **Global Reset**:

Note: To preserve user programs, go to the Reset menu item instead of Global reset. (See Section 8.2.1.)

1. From the Main Menu screen, use the  and  buttons to move the cursor ► to **Maintenance** ►. Press and release the  button.
2. The pump will now display the **Maintenance** submenu screen. The cursor ► will be at **Factory Defaults** ►. Press and release the  button.
3. The cursor ► will now be at **Global reset**. Press and release the  button.
4. The following message box will appear:



5. **Note:** To cancel out of **Global Reset**, press and release the  button or the  button at this time.
6. Press and release the  or  button. The message box now displays:



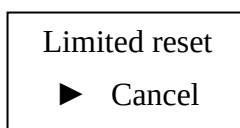
7. Press and release the  button. Global reset is now complete.

8.2.1. Reset (save programs)

Resets all the options as in Global Reset, except that programs are preserved.

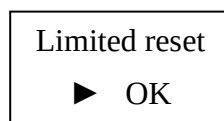
To perform **Reset (save programs)**:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Maintenance** ►. Press and release the  button.
2. The pump will now display the **Maintenance** submenu screen. The cursor ► will now be at **Factory Defaults** ►. Press and release the  button.
3. The cursor ► will now be at **Global reset**. Use the  and  buttons to move the cursor ► to **Reset (save programs)**. Press and release the  button.
4. The following message box will appear:



Note: To cancel out of **Reset (save programs)**, press and release the  button at this time.

5. Press and release the  or  button. The message box now displays:



6. Press and release the  button. Limited reset is now complete.

8.3. T ambient Cal ►

Calibrates the ambient temperature sensor (available in STP version of the product). The calibration should be checked periodically, typically at the beginning of a sample run. Do not exceed six (6) months between calibrations of the ambient temperature sensor.

To perform **T ambient Cal**:

1. The pump should be idle and in a stable ambient temperature for at least one hour before this procedure. This ensures the ambient temperature sensor is not elevated by internal heating.

2. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Maintenance ►**. Press and release the  button. The pump will now display the **Maintenance** submenu screen. The cursor ► will now be at **Factory Defaults ►**.
3. Use the  and  buttons to move the cursor ► to **T ambient Cal ►**. Press and release the  button.
4. The following message box will appear:

Enter Ta (C) 27

Ta Sensor (C) 23

T

Note: To cancel out of **T ambient Cal**, press and release the  button at this time.

5. Press and release the  or  button to adjust the “Enter Ta (C)” value to the current ambient temperature. Press and release the  button. The Ta sensor (C) will change to show the current measured ambient temperature after calibration.
6. Press and release the  button. Ambient temperature calibration is now complete.

8.4. Barometric P Cal ►

Calibrates the barometric pressure sensor (available in STP version of the product). The calibration should be checked periodically, typically at the beginning of a sample run. Do not exceed six (6) months between calibrations of the barometric pressure sensor.

To perform **Barometric P Cal**:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Maintenance ►**. Press and release the  button. The pump will now display the **Maintenance** submenu screen. The cursor ► will now be at **Factory Defaults ►**.
2. Use the  and  buttons to move the cursor ► to **Barometric P Cal ►**. Press and release the  button.

- The following message box will appear:

Enter PB (mmHg)
 PB sensor (mmHg)

Note: To cancel out of **Barometric P Cal**, press and release the  button at this time.

- Press and release the  or  button to adjust the “Enter PB (mmHg)” value to the current barometric pressure. Press and release the  button. The PB (mmHg) will change to show the current measured ambient barometric pressure after calibration.
- Press and release the  button. Barometric pressure calibration is now complete.

8.5. Pressure ►

Calibrates the back pressure sensor. The calibration should be checked periodically, typically at the beginning of a sample run. Do not exceed six (6) months between calibrations of the back pressure sensor.



To perform **Pressure** calibration:

- Set up the GilAir Plus pump, load panel and reference meters as shown. For this procedure, the pump must be set to the **Lo** range of operation. During the calibration, the pump will run and maintain 35 inches of water backpressure. The load valve should be set to allow a flow rate of 100-200 cc/min of flow with this pressure.

2. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Maintenance ►**. Press and release the  button. The pump will now display the **Maintenance** submenu screen. The cursor ► will now be at **Factory Defaults ►**.
3. Use the  and  buttons to move the cursor ► to **Pressure ►**. Press and release the  button.
4. The following message box will appear:

Pressure inH2O 35

After a short delay, the pump will start to run and slowly ramp up power until the pressure is stable at its internally calibrated value. The load valve should be adjusted so that the flow is 100-200 cc/min.

Note: To cancel out of **Pressure**, press and release the  button at this time.

5. Once the GilAir Plus pump is running stable with a flow rate between 100 and 200 cc/min. press and release the  or  button to adjust the “Pressure” value to the pressure sensor reference meter reading. Press and release the  button. A brief “Calibration in Progress” message will appear. The pressure display will return to the original display with the calibration changed. Press and release the  button. The back pressure sensor calibration is now complete.

SECTION NINE: User Maintenance

9.1. Battery Maintenance NiMH

The NiMH battery pack provides service for 300 to 500 cycles depending on the conditions of use and storage.

All rechargeable batteries lose charge over time even if not in use. This is called self discharge. The self discharge rate of the NiMH cells used in the GilAir Plus battery pack is about 1% per day. It is important to have the pack fully charged when starting a sample. To guarantee full charge, leave the pack or pump on the charging dock until ready to use. Once the pump has charged completely, the charging dock will change to a trickle charge mode that maintains the battery at maximum capacity.

If a battery pack goes unused and/or uncharged for more than three (3) weeks, it should be reconditioned by putting it on the charging dock for 72 hours. The dock will fast charge the battery and the additional trickle charge time will recondition it for maximum capacity.

As the battery goes through charge/discharge cycles and ages, its capacity will diminish. If the battery is reconditioned as described above and does not result in expected run times, the battery has reached end of life and should be replaced.

Battery life can be extended by storing in cool conditions. High temperatures can cause accelerated aging.

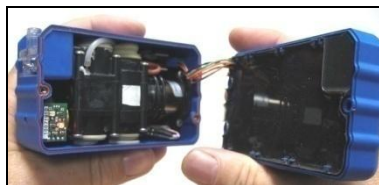
9.2. Battery Replacement

To replace the rechargeable NiMH battery pack, follow the steps below:

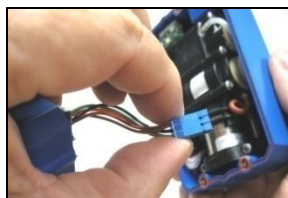
1. Remove the three case screws.
2. Remove the battery pack from the front case. Note that it is connected to the PC board by a wire harness.
3. Carefully unplug the wire harness, noting the position of the six pin connector.
4. Connect the new battery, observing the correct pin pattern.
5. Reconnect the case halves. Pay attention to wire routing so as not to pinch the wire harness.
6. Replace the case screws, and snug tighten only. Tighten until the gap between the case halves fully closes – **Do not over tighten.**



Step 1



Step 2



Steps 3-4

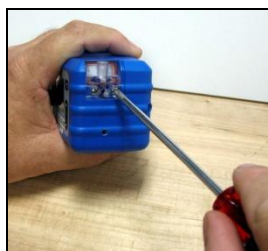


Steps 5-6

9.3. Pump Filter Maintenance

If the flow resistance of the input filter is too high, the sampling load capacity of the pump will be reduced. The status of the filter is easily determined. If the pressure drop across the filter is higher than 2 inches of water at a flow rate of 5000 cc/min, it should be replaced. To make this measurement, set the pump flow rate to 5000 cc/min running in constant flow mode. With nothing hooked to the inlet connector, the back pressure displayed on the pump should be less than 2 inches of water. If the Back Pressure is larger, replace the filter using the procedure below.

1. Remove the two screws from the filter housing.
2. Remove the filter housing by pulling straight out from the pump case.
3. Replace the filter P/N 811-0905-01R . Be sure it is properly positioned and gasket seal areas are clear (Gasket P/N 300-0103-01R). Insert the filter housing in the manifold. Note the proper positioning of the two o-rings, P/N 150-9106-50 R and 150-9121-50 R (Replace if damaged).
4. Re-attach the filter housing. Align outer gasket area first. Align filter housing exit connection and press inward until the o-rings are seated. Be sure the filter housing is fully seated before tightening screws.
5. Replace screws and snug tighten only. Do not over-tighten.



Step 1



Step 2



Step 3



Steps 4-5

SECTION TEN: Appendices

Appendix A: Menu Outline

Underlined option indicates factory default (shipped) settings

RUN

FLOW SET

CALIBRATE

SETUP ▶

- ▶ **EVENT ID ENABLE** (enable / disable)
- ▶ **PRE/POST CAL** (enable / disable)
- ▶ **FAULT RETRY** (enable / disable)
- ▶ **VALVE MODE** (continuous / start/stop)
- ▶ **SmartCal** (Manual / Gilibrator or other than approved by Sensidyne)
- ▶ **CLEAR DATALOG**
- ▶ **RUN OPTIONS ▶**
 - ▶ **STANDARD TEMP**
 - ▶ **STANDARD PRESSURE**
- ▶ **DISPLAY OPTIONS ▶**
 - ▶ **LANGUAGE** (English / Spanish / French / Dutch / German / Italian / Portuguese)
 - ▶ **TEMPERATURE UNITS** (°C / °F)
 - ▶ **PRESSURE UNITS** ("H₂O / mmHg / KPa)
- ▶ **CLOCK SET ▶**
 - ▶ **CLOCK** (hours / minutes)
 - ▶ **DATE** (year / month / date)
 - ▶ **TIME FORMAT** (12 hour / 24 hour)
 - ▶ **DATE FORMAT** (mm/dd/yy / dd/mm/yy)
- ▶ **PASSWORD** (0)

CONTROL MODE (CE / CPL / CPH)

RUN MODE (Manual / Timed / PROG #)

RUN SETUP ▶

- ▶ **TIMED START** (00:00 / 00:00)
- ▶ **TIMED DURATION** (01)
- ▶ **PRESS SET** (" H₂O) (18.0)
- ▶ **PROGRAM** ▶
 - ▶ **ENABLE** (disable / enable)
 - ▶ **PROGRAM EDIT** ▶
 - ▶ **PROG NAME** (PROG 1)
 - ▶ **CONTROL MODE** (CE / CPL / CPH)
 - ▶ **SETPOINT** (2000)
 - ▶ **PROG STEP** (01)
 - ▶ **FUNCTION** (Time / On Interval / Off Interval / Cycle)
 - ▶ **FUNCTION VALUE** 00:00
 - ▶ **SAVE**

REVIEW

MAINTENANCE ▶

- ▶ **FACTORY DEFAULTS** ▶
 - ▶ **GLOBAL RESET**
 - ▶ **RESET (SAVE PROGRAMS)**
- ▶ **T-AMBIENT CAL**
 - Enter Ta (°C) --
 - Ta Sensor (°C) --
- ▶ **BAROMETRIC P CAL**
 - Enter PB (mmHg) ---
 - PB Sensor (mmHg) ---
- ▶ **PRESSURE**
 - Pressure in " H₂O ---
 - PB Sensor (mmHg) ---

Appendix B: Example Program Setup & Edit

This example program operates the pump as follows:

Constant Flow Mode.

3000 cc/min flow rate calibrated flow rate.

The program starts with an On Interval at 4:00 PM (16:00 Hrs).

The On Interval lasts for 1 minute, then an Off Interval begins and the pump stops running.

The Off Interval lasts for 1 minute, then repeats the entire program for 2 cycles.

The pump is calibrated per Section 4.9.2. and 4.9.3. to run at a flow rate of 3000 cc/min.

ENTER THE PROGRAM

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Setup** ►. Press and release the  button. Use the  and  buttons to move the cursor ► to **Program** ►. Press and release the  button.
2. The cursor ► is now at **Enable**. Use the  and  buttons to set Enable to enable. Press and release the  button.
3. Use the  and  buttons to move the cursor ► to Program Edit ►. Press and release the  button.
4. The cursor ► is now at **Prog Name**. Use the  and  buttons to enter 1 for the program name. Press and release the  button. PROG 1 now appears to the right of Prog Name.
5. Use the  and  buttons to move the cursor ► to **Control Mode**. Use the  and  buttons to set the Control Mode to CF (Constant Flow). Press and release the  button.
6. Use the  and  buttons to move the cursor ► to **Setpoint**. Use the  and  buttons to set the set point flow rate to 3000 cc/min. Press and release the  button.
7. Use the  and  buttons to move the cursor ► to **Prog Step**. Use the  and  buttons to set the program step to 01. Press and release the  button.

8. Use the  and  buttons to move the cursor ► to **Function**. Use the  and  buttons to set the function to **Time**. Press and release the  button.
9. Use the  and  buttons to move the cursor ► to **Function value**. Use the  and  buttons to set the hour to 16:00. Press and release the  button. The minutes are now underlined. Use the  and  buttons to set the minutes to 00. Press and release the  button. 16:00 is now the displayed program start time.
10. Use the  and  buttons to move the cursor ► back up to **Prog Step**. Use the  and  buttons to set the program step to 02. Press and release the  button.
11. Use the  and  buttons to move the cursor ► to **Function**. Use the  and  buttons to set the function to On Interval. Press and release the  button.
12. Use the  and  buttons to move the cursor ► to **Function value**. Use the  and  buttons to set the On Interval to 01. Press and release the  button.
13. Use the  and  buttons to move the cursor ► back up to **Prog Step**. Use the  and  buttons to set the program step to 03. Press and release the  button.
14. Use the  and  buttons to move the cursor ► to **Function**. Use the  and  buttons to set the function to Off Interval. Press and release the  button.
15. Use the  and  buttons to move the cursor ► to **Function value**. Use the  and  buttons to set the Off Interval to 01. Press and release the  button.
16. Use the  and  buttons to move the cursor ► back up to **Prog Step**. Use the  and  buttons to set the program step to 04. Press and release the  button.
17. Use the  and  buttons to move the cursor ► to **Function**. Use the  and  buttons to set the function to Cycle. Press and release the  button.
18. Use the  and  buttons to move the cursor ► to **Function value**. Use the  and  buttons to set the Off Interval to 02. Press and release the  button.

19. Use the  and  buttons to move the cursor ► to **Save**. Press and release the  button.
20. Press the  button 3 times to exit to the Main Menu. Use the  and  buttons to move the cursor ► to Run Mode. Use the  and  buttons to set the Run Mode to PROG 1. Press and release the  button.
21. Use the  and  buttons to move the cursor ► to **Run**. Press and release the  button. The pump will then display the Event ID screen (if the Event function has been enabled see Section 5.2.). If Event has been enabled, press and release the  button. The pump now displays the Program screen, and the pump will begin to run at the displayed start time (16:00).

EDITING THE PROGRAM

To change the number of cycles in the program from 2 cycles to 3 cycles, use the following steps:

1. At the Main Menu screen, use the  and  buttons to move the cursor ► to **Run Setup ►**. Press and release the  button. Use the  and  buttons to move the cursor ► to **Program ►**. Press and release the  button.
2. Use the  and  buttons to move the cursor ► to **Program Edit ►**. Press and release the  button.
3. The cursor ► is now at **Prog Name**. Use the  and  buttons to enter 1 for the program name. Press and release the  button. PROG 1 now appears to the right of Prog Name.

IMPORTANT! – To edit an already existing program, you must correctly enter the program name of the program you wish to edit. Otherwise, you may inadvertently edit the wrong program!

In this example, only one program was entered and stored into the pump. When the program was entered, it was assigned the unique program name PROG 1 (any other name could have been assigned as well, i.e. – PROG 3). Enter PROG 1 to edit the program with the unique name PROG 1.

Note: Unique program identifiers can be assigned via the PC Program Manager.

To change the number of cycles in Program 1, access the Program Step where the Cycle function was entered. For the existing program, the Cycle function was selected at Program Step 04.

1. Use the  and  buttons to move the cursor ► to **Prog Step**. Use the  and  buttons to set the program step to 04. Press and release the  button.
2. Use the  and  buttons to move the cursor ► to **Function**. Verify that the setting assigned to Function for Program Step 4 is Cycle.
3. Use the  and  buttons to move the cursor ► to **Function value**. Use the  and  buttons to change the Function value from 02 to 03. Press and release the  button.
4. Use the  and  buttons to move the cursor ► to **Save**. Press and release the  button. The change from 2 cycles to 3 cycles for the program named PROG 1 has now been successfully entered and stored. Press and release the  button 3 times to exit to the Main Menu.

Appendix C: Troubleshooting Guide

Symptom	Possible Cause	Corrective Action
Run time too short; does not meet projected run time	Battery Pack not fully charged.	Charge battery (Sec. 3.1.)
	Batteries have discharged. Battery pack needs to be reconditioned.	Recondition battery pack (Sec. 9.1.)
	Battery pack capacity too low, at end of life	Replace battery (Sec. 9.2.)
Pump will not turn on	Low battery charge	Charge battery (Sec. 3.1.)
	Blown fuse in battery	Replace battery (Sec. 9.2.)
	Dead Cell in battery	Replace battery (Sec.9.2.)
	Control board problem	Return for service
Pump shows Fault in display/Enters HOLD	Inlet filter clogged	Replace Filter (Sec. 9.3.)
	Intake obstructed	Examine sample holder and remove obstruction or run at lower flow rate
	Control board problem	Return for service
	Valve change during operation	Valve cannot be changed during sample, only before starting event
	Valve error shown, no valve change attempted	Valve may be affected by strong magnetic fields (Sec. 5.5.)
	Low battery	Charge battery (Sec. 3.1.)
	Flow rate is set too high for sample media	Correct the flow rate per sample method guidelines
	Sample media tubing pinched shut	Correct tubing obstruction
Pump runs flat out	Internal flow transducer problem	Back pressure too high; remove restriction or lower flow rate
	Control board problem	Return for service
	Improper Calibration	Recalibrate Pump (Sec. 4.9.3.)

Symptom	Possible Cause	Corrective Action
Pump runs erratic & faults	Faulty bearing	Return for service
	Faulty motor	Return for service
	Liquid in pump or other matter	Return for service
	Charger connected	Don't run pump with charger connected
	Insufficient back pressure for stable operation	Check sample media or increase flow rate
Pump does not achieve flow specifications	Valve dirty or torn	Return for service
	Torn diaphragm on yoke assembly	Return for service
	Leak in pump	Input manifold screws may not be tight after replacing input filter. Tighten screws to snug tight only. Do not over tighten. Return for service if tightening screws does not solve leakage issue. (Sec. 9.3.)
	Battery not sufficiently charged	Charge battery (Sec. 3.1.)
Pump surges	Display calibration adjusted out of range	Reset display calibrations to factory setting (Sec. 8.2.)
	Charger connected	Don't run pump with charger connected
Pump will not run program	Program time is set to zero	Enter non zero program duration (Sec. 6.1.)
	Program not enabled	Check program enable (Sec. 6.2.)
	Program not selected in run mode	Check run setup (Sec. 5.14.)
	Not DL or STP Model	Upgrade to DL or STP model
Keypad inoperative	Keypad is locked	Unlock Keypad (Sec. 4.7.3.)
	Pump in off phase of program	Wait for program to complete program or stop program

Symptom	Possible Cause	Corrective Action
Pump stops occasionally and restarts after 15 seconds	Normal self-check function. Flow control is being re-zeroed.	None (Sec. 4.6.)
Displayed flow rate does not match calibration	Flow display is out of calibration	Calibrate flow display (Sec. 4.9.3.)
Pump will not run at desired flow rate in low flow mode	Wrong sample tube holder selected for constant pressure control adapter	Select tube holder that incorporates a needle valve
Pump does not start, "Change Valve" displayed	Flow valve not in correct mode	Reset valve (Sec. 4.8.1.)
"Check Device" displayed	Device communication error in SmartCal	Check that calibration device is on and cable is connected (Sec. 5.6.)
Pump continuously displays 'NIMH analysis' and steady red LED for long periods, with intermittent off period	Battery has deeply discharged due to lack of use or is damaged	Leave pump on charge for two hours. If red fast charge or green LED does not appear replace battery pack.

Appendix D: Parts List

Part Number	Description
810-0901-01-R	Basic Pump Only (No Dock)
810-0902-01-R	Datalogging Pump Only (No Dock)
810-0903-01-R	STP Pump Only (No Dock)
910-0901-US-R	GilAir Plus Basic Single Starter Kit, US cord
910-0901-EU-R	GilAir Plus Basic Single Starter Kit, Euro cord
910-0901-UK-R	GilAir Plus Basic Single Starter Kit, UK cord
910-0902-US-R	GilAir Plus Datalogging Single Starter Kit, US cord
910-0902-EU-R	GilAir Plus Datalogging Single Starter Kit, Euro cord
910-0902-UK-R	GilAir Plus Datalogging Single Starter Kit, UK cord
910-0903-US-R	GilAir Plus STP Single Starter Kit, US cord
910-0903-EU-R	GilAir Plus STP Single Starter Kit, Euro cord
910-0903-UK-R	GilAir Plus STP Single Starter Kit, UK cord
910-0907-US-R	GilAir Plus Basic 3-Pump Starter Kit, US cord
910-0907-EU-R	GilAir Plus Basic 3-Pump Starter Kit, Euro cord
910-0907-UK-R	GilAir Plus Basic 3-Pump Starter Kit, UK cord
910-0908-US-R	GilAir Plus Datalogging 3-Pump Starter Kit, US cord
910-0908-EU-R	GilAir Plus Datalogging 3-Pump Starter Kit, Euro cord
910-0908-UK-R	GilAir Plus Datalogging 3-Pump Starter Kit, UK cord
910-0909-US-R	GilAir Plus STP 3-Pump Starter Kit, US cord
910-0909-EU-R	GilAir Plus STP 3-Pump Starter Kit, Euro cord
910-0909-UK-R	GilAir Plus STP 3-Pump Starter Kit, UK cord

Part Number	Description
910-0904-US-R	GilAir Plus Basic 5-Pump Starter Kit, US cord
910-0904-EU-R	GilAir Plus Basic 5-Pump Starter Kit, Euro cord
910-0904-UK-R	GilAir Plus Basic 5-Pump Starter Kit, UK cord
910-0905-US-R	GilAir Plus Datalogging 5-Pump Starter Kit, US cord
910-0905-EU-R	GilAir Plus Datalogging 5-Pump Starter Kit, Euro cord
910-0905-UK-R	GilAir Plus Datalogging 5-Pump Starter Kit, UK cord
910-0906-US-R	GilAir Plus STP 5-Pump Starter Kit, US cord
910-0906-EU-R	GilAir Plus STP 5-Pump Starter Kit, Euro cord
910-0906-UK-R	GilAir Plus STP 5-Pump Starter Kit, UK cord
811-0901-US-R	Single Station Basic Dock 100-240Vac, 50-60 Hz, US Cord (includes Power Supply)
811-0901-EU-R	Single Station Basic Dock 100-240Vac, 50-60 Hz, Euro Cord (includes Power Supply)
811-0901-UK-R	Single Station Basic Dock 100-240Vac, 50-60 Hz, UK Cord (includes Power Supply)
811-0911-US-R	Three Station Basic Dock 100-240Vac, 50-60 Hz, US Cord. (includes Power Supply)
811-0911-EU-R	Three Station Basic Dock 100-240Vac, 50-60 Hz, Euro Cord. (includes Power Supply)
811-0911-UK-R	Three Station Basic Dock 100-240Vac, 50-60 Hz, UK Cord. (includes Power Supply)
811-0902-US-R	Five Station Basic Dock 100-240Vac, 50-60 Hz, US Cord. (includes Power Supply)
811-0902-EU-R	Five Station Basic Dock 100-240Vac, 50-60 Hz, Euro Cord. (includes Power Supply)
811-0902-UK-R	Five Station Basic Dock 100-240Vac, 50-60 Hz, UK Cord. (includes Power Supply)

Part Number	Description
811-0903-US-R	Single Station Dock w/Comms 100-240Vac, 50-60 Hz, US Cord (includes Power Supply)
811-0903-EU-R	Single Station Dock w/Comms 100-240Vac, 50-60 Hz, Euro Cord (includes Power Supply)
811-0903-UK-R	Single Station Dock w/Comms 100-240Vac, 50-60 Hz, UK Cord (includes Power Supply)
811-0912-US-R	Three Station Dock w/Comms 100-240Vac, 50-60 Hz, US Cord (includes Power Supply)
811-0912-EU-R	Three Station Dock w/Comms 100-240Vac, 50-60 Hz, Euro Cord (includes Power Supply)
811-0912-UK-R	Three Station Dock w/Comms 100-240Vac, 50-60 Hz, UK Cord (includes Power Supply)
811-0904-US-R	Five Station Dock w/Comms 100-240Vac, 50-60 Hz, US Cord (includes Power Supply)
811-0904-EU-R	Five Station Dock w/Comms 100-240Vac, 50-60 Hz, Euro Cord (includes Power Supply)
811-0904-UK-R	Five Station Dock w/Comms 100-240Vac, 50-60 Hz, UK Cord (includes Power Supply)
811-0908-US-R	Power Supply, Single Station 100-240Vac, 50-60 Hz, US Cord
811-0908-EU-R	Power Supply, Single Station 100-240Vac, 50-60 Hz, EU Cord
811-0908-UK-R	Power Supply, Single Station 100-240Vac, 50-60 Hz, UK Cord
811-0909-US-R	Power Supply, 3/5 -Station 100-240Vac, 50-60 Hz, US Cord
811-0909-EU-R	Power Supply, 3/5 -Station 100-240Vac, 50-60 Hz, EU Cord
811-0909-UK-R	Power Supply, 3/5 -Station 100-240Vac, 50-60 Hz, UK Cord
780-0015-01-R	Calibrator cable (Gilibrator), for SmartCal option
780-0015-02-R	Calibrator cable (Challenger®), for SmartCal option
780-0015-03-R	Calibrator cable (TSI™), for SmartCal option
780-0015-04-R	Calibrator cable (BIOS Defender™), for SmartCal option

Part Number	Description
783-0012-01-R	Battery Pack NiMH Rechargeable
783-0013-01-R	Battery Pack AA Alkaline (Primary Cells)
783-0014-01-R	Battery Pack Bench Top DC Power Adapter (for area sampling only).
811-0907-01-R	Dock to PC, USB cable (w/inductor), for Com version dock only
811-0905-01-R	Inlet Filters, Replacement, Pack of 10
360-0132-01	Operation Manual
360-0135-01	Quick Start Guide
360-0143-01	PC Application Manual
811-0913-01-R	Outlet Fitting, Bag Fill Adapter
811-0914-01-R	Tripod Mount, Adapts pump to camera style tripod, GilAir/GilAir Plus/3500/5000
800573-3	Diagnostic Panel
800783-3	Diagnostic Panel with Stand
800143	Filter Cassette Holder Kit
800149	Single Tube Holder Kit, 6 x 70, for 150 mg Activated Charcoal tube
297-0006-01-R	Power Cord, Dock, US
297-0007-01-R	Power Cord, Dock, Euro
297-0008-01-R	Power Cord, Dock, UK

Appendix E: Specifications

PERFORMANCE

Flow Range..... 20 - 5000 cc/min (450 - 5000 cc/min in Constant Flow High Flow mode, 20-449 cc/min in Constant Flow Low Flow mode)
1-5000 cc/min in Constant Pressure mode

Flow Modes..... High and Low Flow Constant Flow or Constant Pressure

Flow Display..... $\pm 5\%$ of set flow or ± 3 cc/min whichever is higher

Constant Flow Control..... $\pm 5\%$ of set flow or ± 3 cc/min whichever is higher from minimum pressure of 2 "H2O to specified maximum

Constant Pressure Control... $\pm 10\%$ of back-pressure

Run Time/Back Pressure Capability	Flow Range	BP for 8 Hrs	Fault
	5000 cc/min	12" H2O	minimum 15" H2O
	4000 cc/min	20" H2O	minimum 30" H2O
	3000 cc/min	30" H2O	minimum 35" H2O
	2000 cc/min	30" H2O	minimum 40" H2O
	1000 cc/min	35" H2O	minimum 40" H2O
	450-999 cc/min	40" H2O	minimum 40" H2O
	20-449 cc/min	25" H2O	minimum 25" H2O

Flow Fault..... If flow changes exceed 5% within back pressure specifications, fault notification appears. If fault exceeds 30 seconds, pump shuts down. Selectable automatic fault recovery allows the pump to attempt restart every 3 minutes for up to 10 times or to hold until manual intervention.

ENVIRONMENTAL

Temperature Ranges

Operating.....32°F to 113°F (0°C to 45°C)

Storage.....-4°F to 113°F (-20°C to 45°C)

Charging..... 41°F to 104°F (5°C to 40°C)

Humidity Ranges

Operating..... 5-95 %RH, non-condensing

Storage..... 5-98 %RH, non-condensing

GENERAL

RoHS Compliant.....European and China, without taking advantage of
product use exemptions listed for use in both
unclassified and hazardous classified areasDisplay.....Flow rate, sample time, and sample volume in actual
conditions

Indicators.....Green and Red LED's

Dimensions.....4.3W x 2.4H x 2.4D inches (11.0W x 6.1H x 6.1D cm)

Weight.....20.5 oz. (580 g)

Hex Key.....2 mm or 5/64 inch hex

ELECTRICAL

Power Options..... Removable rechargeable nickel-metal hydride (NiMH)
battery pack, optional alkaline battery pack, or DC
Power

Battery Level Indicator..... On-screen icon displays battery level

Interface Connectors.....All pump computer interface performed via the Dock

Dock Functions.....Battery charging, USB computer interface (optional),
and SmartCal automatic-calibration (optional)

Charging Time.....3.5 hours or less

Appendix F: Charging/Communications Dock

The Dock provided with the GilAir Plus charges the pump and optionally provides communications capability for a computer connection and connection to an air flow calibrator for automatic calibration (i.e., SmartCal).

The Dock is available in one, three or five station basic or communication configurations. The basic configuration provides pump charging capability only and does not allow for communications to a PC or air flow calibrator with the pump. The communications configuration enables a USB connection to a PC and use of the GilAir Plus PC Application and the connection of an air flow calibrator device. Each supported air flow calibrator device has a unique adapter cable that must be ordered separately. The devices supported and the part numbers of the communications cables are listed in Section 4.9.2. and in the part number table in Appendix D. See Section 5.6. for further details of the SmartCal feature.

Power is supplied to the Dock through a universal input power supply (100-240 Vac, 50/60Hz) or a car accessory plug. There is no power switch on the Dock as the Dock detects the pump or battery pack when it is placed on the Dock and begins the charging process automatically.

When the NiMH battery pack is used the pump cannot be run while on the Dock. The pump will not start on the Dock with a NiMH battery pack. When using the replaceable cell pack or the DC power pack, the pump can be used while mounted on the Dock.

The charging process begins as soon as the pump with NiMH battery pack is placed on the Dock. The pump initially displays a charging status screen showing 'NiMH analysis'. (Note that this message does not appear if the pump is not powered on when mounted on the Dock.) When the 'NiMH analysis' process confirms a good battery pack, the Dock will begin to fast charge the battery pack. The red LED on the pump flashes at a fast rate to indicate that fast charging is in process. If the 'NiMH analysis' process determines that the battery pack is bad for some reason, the red LED will turn off for a few seconds and the 'NiMH analysis' process will repeat itself. If the battery pack does not pass the 'NiMH analysis' because the batteries are determined to be in a state of deep discharge, the repeated 'NiMH analysis' charge process will recondition the battery pack and the fast charge will start. Note, if the battery pack persists in repeated analysis mode for more than one hour it should be replaced.

Once fast charge begins, the pump will take between 10 minutes and 3.5 hours to complete the main charge operation, depending on the initial charge status of the battery pack. When full charge is detected, the Dock switches to a "top off" charge at a lower charging rate for 25 minutes. This "top off" charge sequence ensures a uniform full charge in all cells in the battery pack. During the "top off" phase the green LED flashes slowly. At this point, the pump is at nominal full charge and can be removed and used, although it is recommended to wait for the "top off" charge sequence to finish before using the pump to ensure maximum run time. After the "top off" phase completes, the green LED turns to steady green and the Dock continues to charge the battery pack at a trickle charge rate that will continue to hold the battery cells at full charge while the battery pack is on the Dock. The battery pack may be removed and used at any time when the LED is green.

Note that if the pump or battery pack remains unused for an extended period of about one month or longer, the battery pack should be reconditioned for maximum run time by leaving the pump or battery pack on the Dock in trickle charge mode (indicated by a steady green LED) for a period of at least 72 hours. This will return all of the cells in the battery pack to a uniform full charge condition.

LED Indication	Charge Stage
Short, occasional red flashes	Waiting for pump or battery to be inserted
Continuous red	Analyzing battery
Fast flashing red	Fast charge
Flashing green	Top off charge
Continuous green	Trickle charge
Long red, short off	Repeated analysis cycles indicating deeply discharged or damaged battery pack

Note that NiMH battery packs may be charged separately on the Dock without being connected to a pump if desired. The same cycle is used as described previously and the LED signals are provided by the LED on the Dock.

The USB connection can be used to interface the pumps held on the Dock to a PC for use with the optional GilAir Plus PC Application. This application transfers event data to a PC database, provides pump management capability including the ability to copy pump setups and allows sampling plans (programs) to be created to control the timing and setting of pump operation. The PC Application is described more fully in the separate GilAir Plus PC Application Manual (P/N 360-0143-01).

Appendix G: Factory Calibration and Service

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